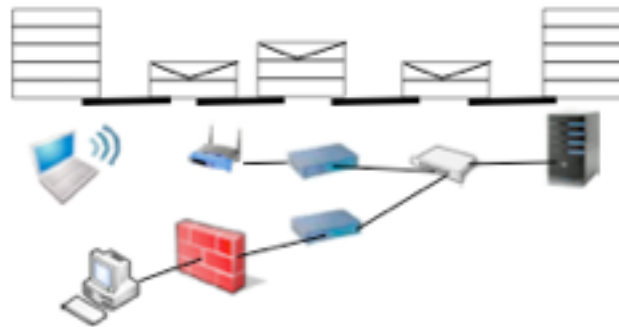

Computer Networking

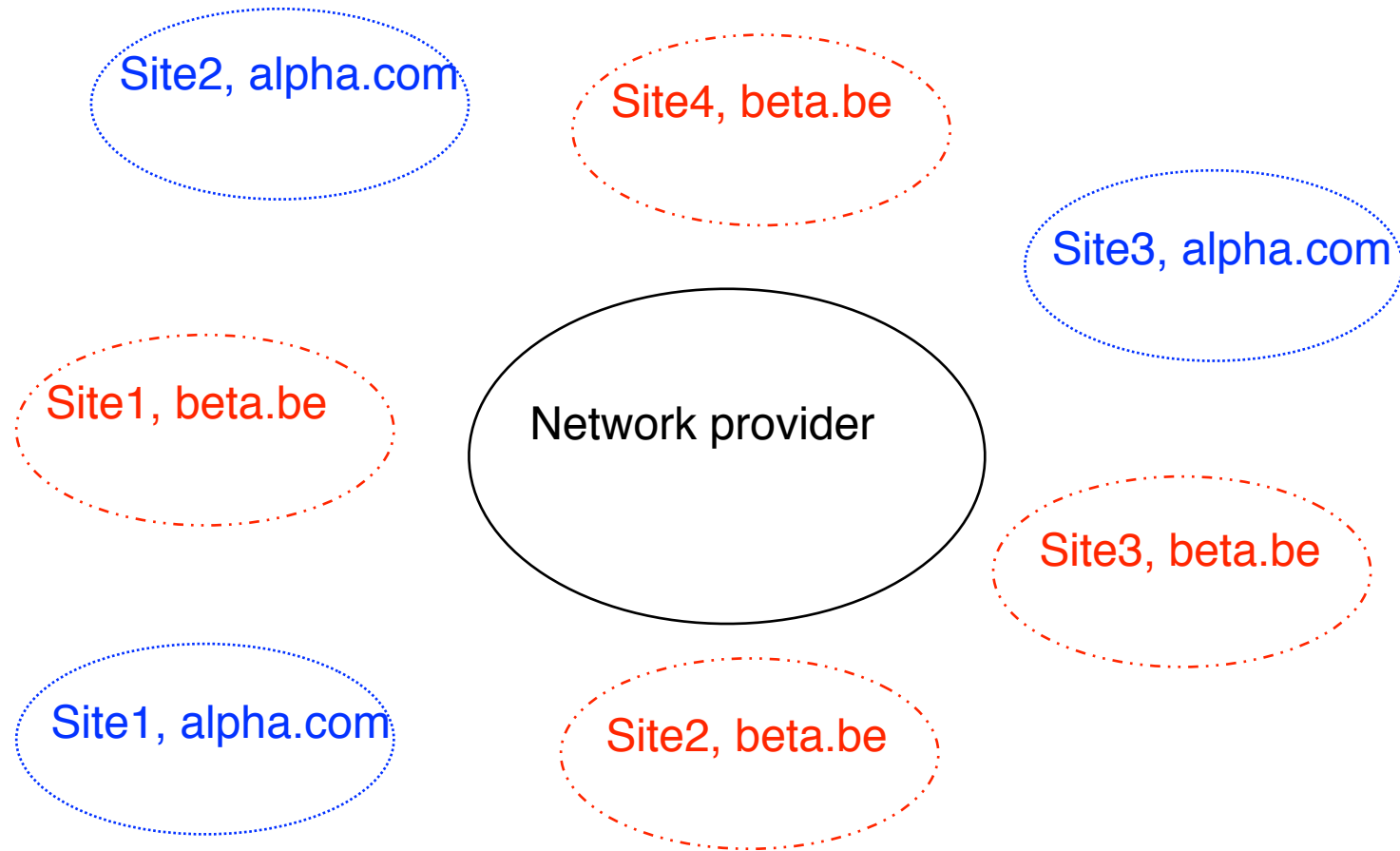
Principles
Protocols
and
Practice



Part 8: Virtual Private Networks

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<http://inl.info.ucl.ac.be/>

The VPN problem

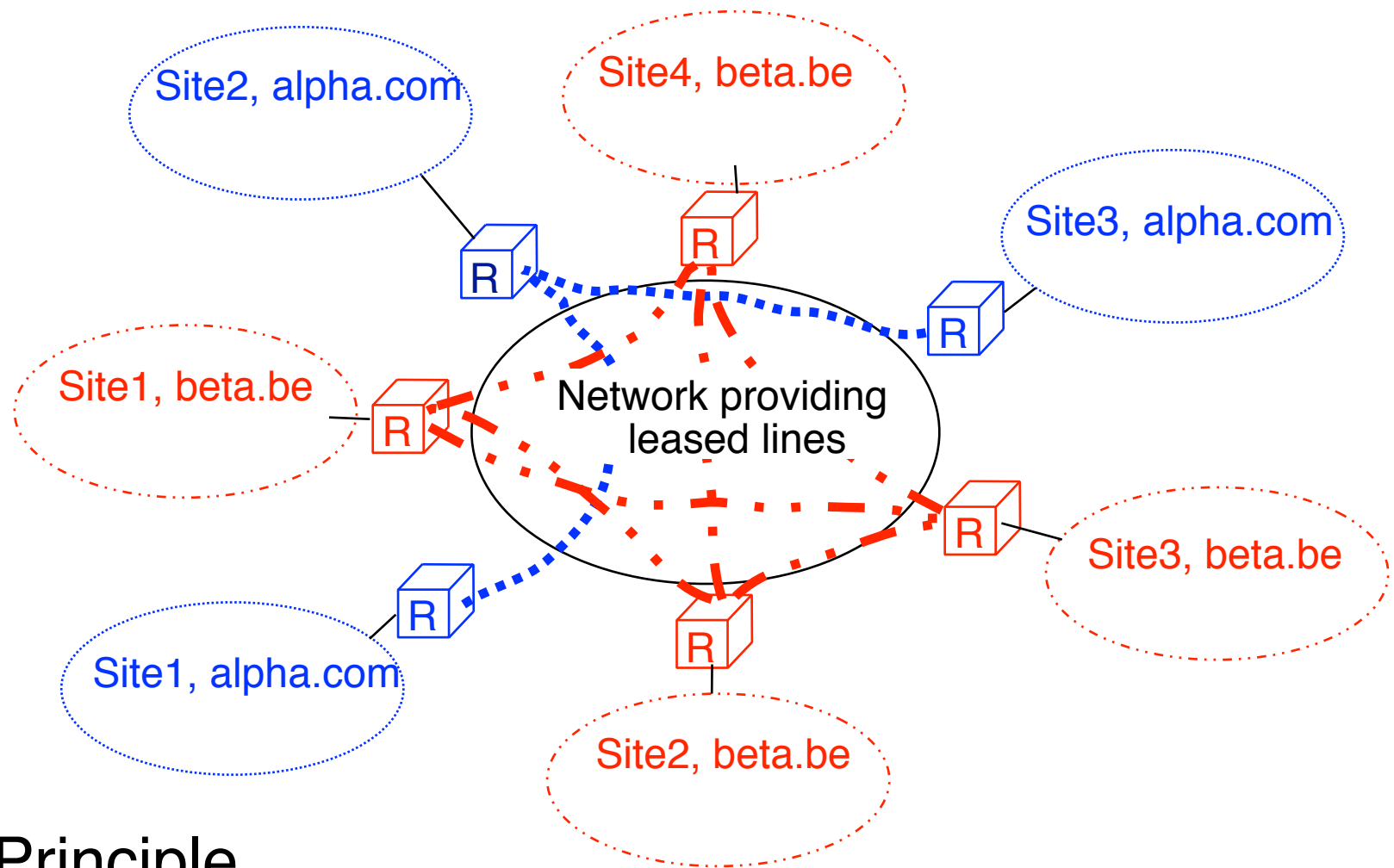


- How to efficiently create
 - one network containing the sites from [alpha.com](#)
 - one network containing the sites from beta.be

What should be the goal of a good VPN ?

- A good VPN service should
 - Support multiple corporate customers
 - in this case, the traffic from these customers should be isolated
 - some security features should be supported to ensure that packets from public Internet can be introduced inside VPN
 - provide QoS guarantees for corporate customers
 - typical solution is to reuse the classical mechanisms
 - be easy to use and manage
 - from the customer viewpoint
 - from the service provider viewpoint

The classical solution

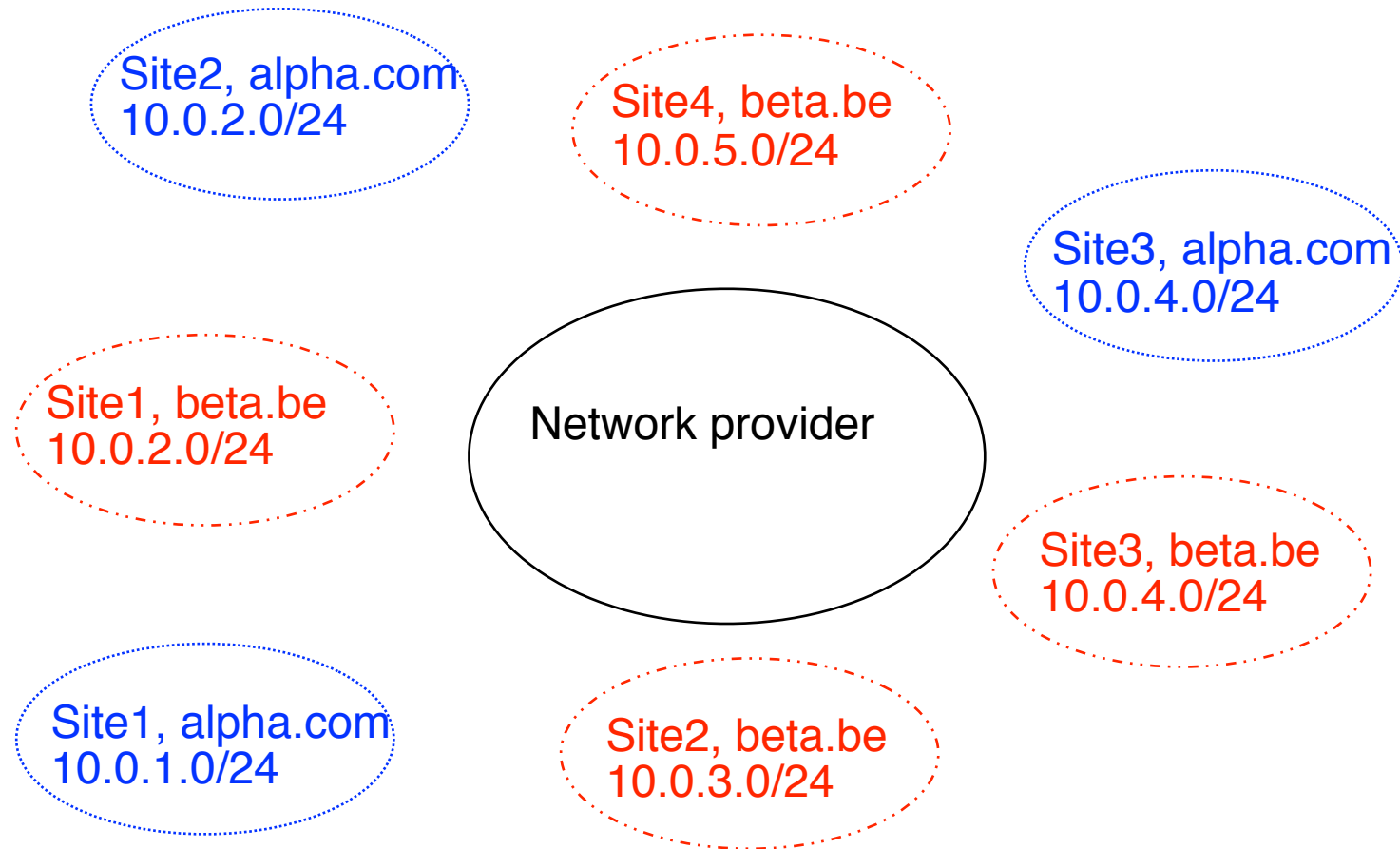


- Principle
 - Create leased lines between sites
 - full mesh (beta.be), hub and spoke (alpha.com) topologies

Evaluation of the classical solution

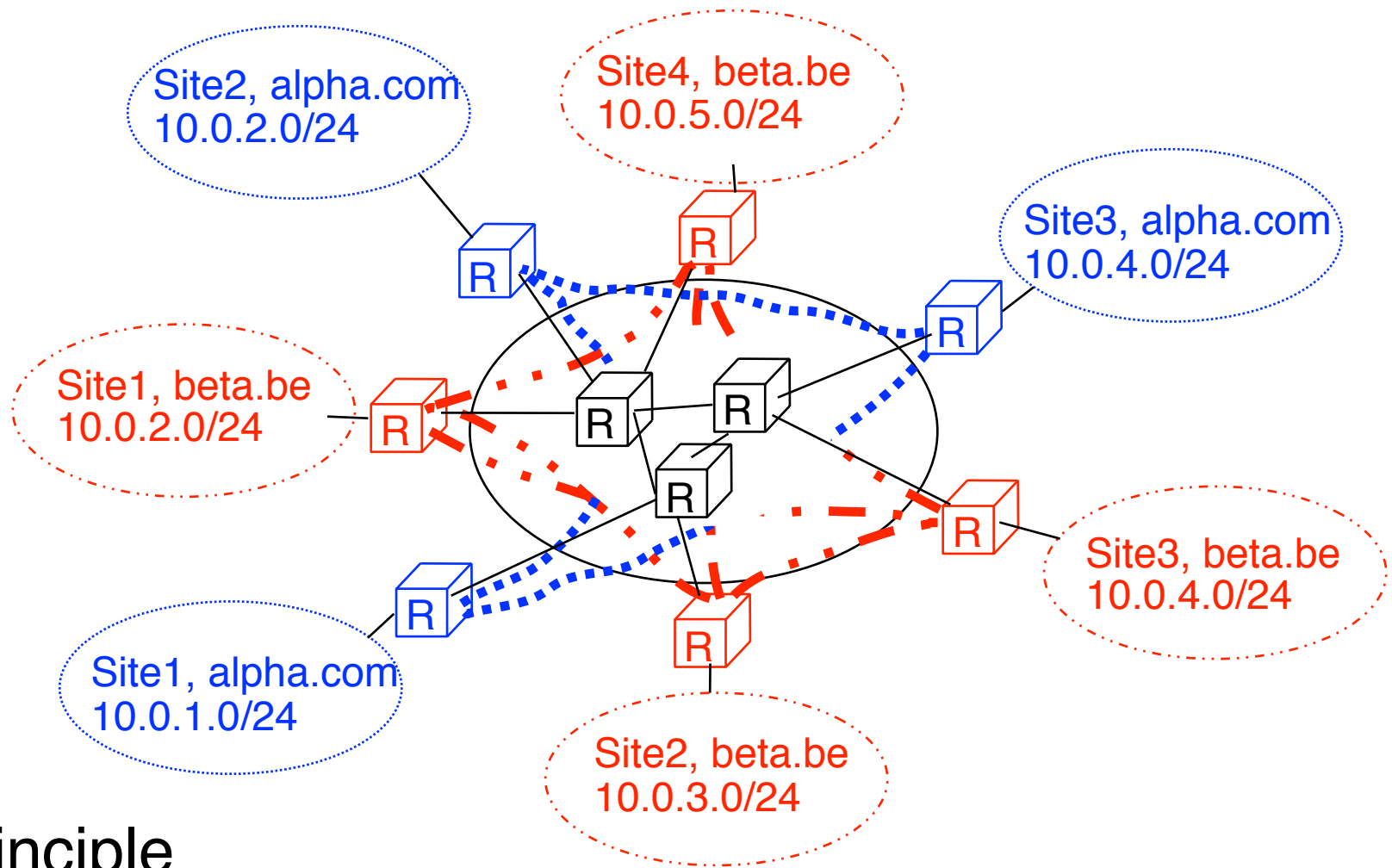
- Advantage
 - the quality of the service provided by the service provider is usually very good
- Drawbacks
 - the number of leased lines can be high
 - $n*(n-1)/2$ leased lines in total for full mesh
 - For a VPN with n sites, each router needs $n-1$ interfaces to obtain a full mesh
 - Flexibility
 - addition of a VPN may require several new lines
 - installation of leased line may require $O(\text{months})$
 - cost can be high
 - no statistical multiplexing on provider's backbone
 - link costs even if no traffic is exchanged

The IP-VPN problem



- How to efficiently create
 - one network containing the sites from [alpha.com](#)
 - one network containing the sites from beta.be

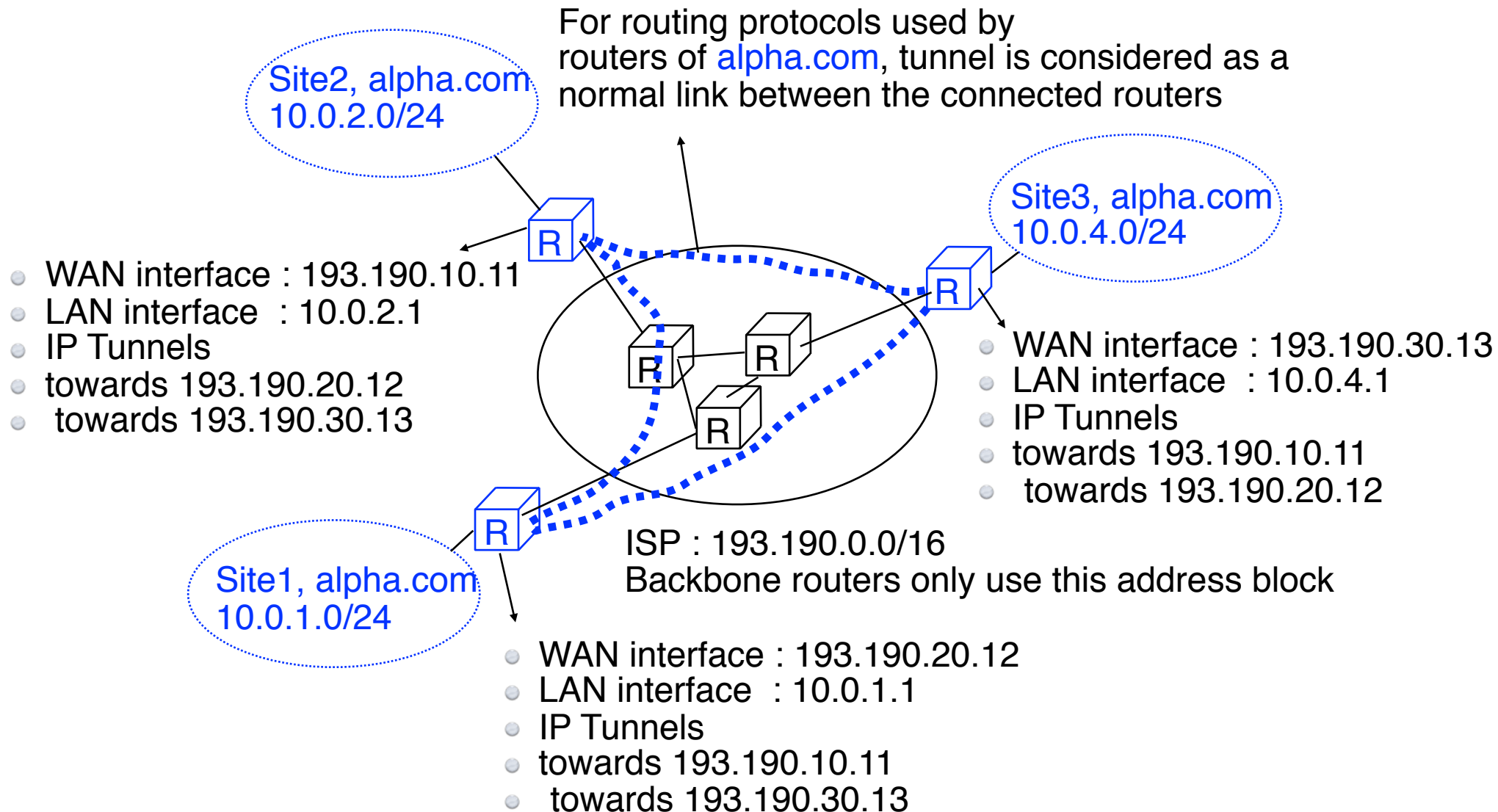
A customer-provisioned IP VPN



- Principle

- create IP tunnels from customer routers through ISP
- drawback : configuration burden on customer routers

A customer-provisioned IP-VPN (2)

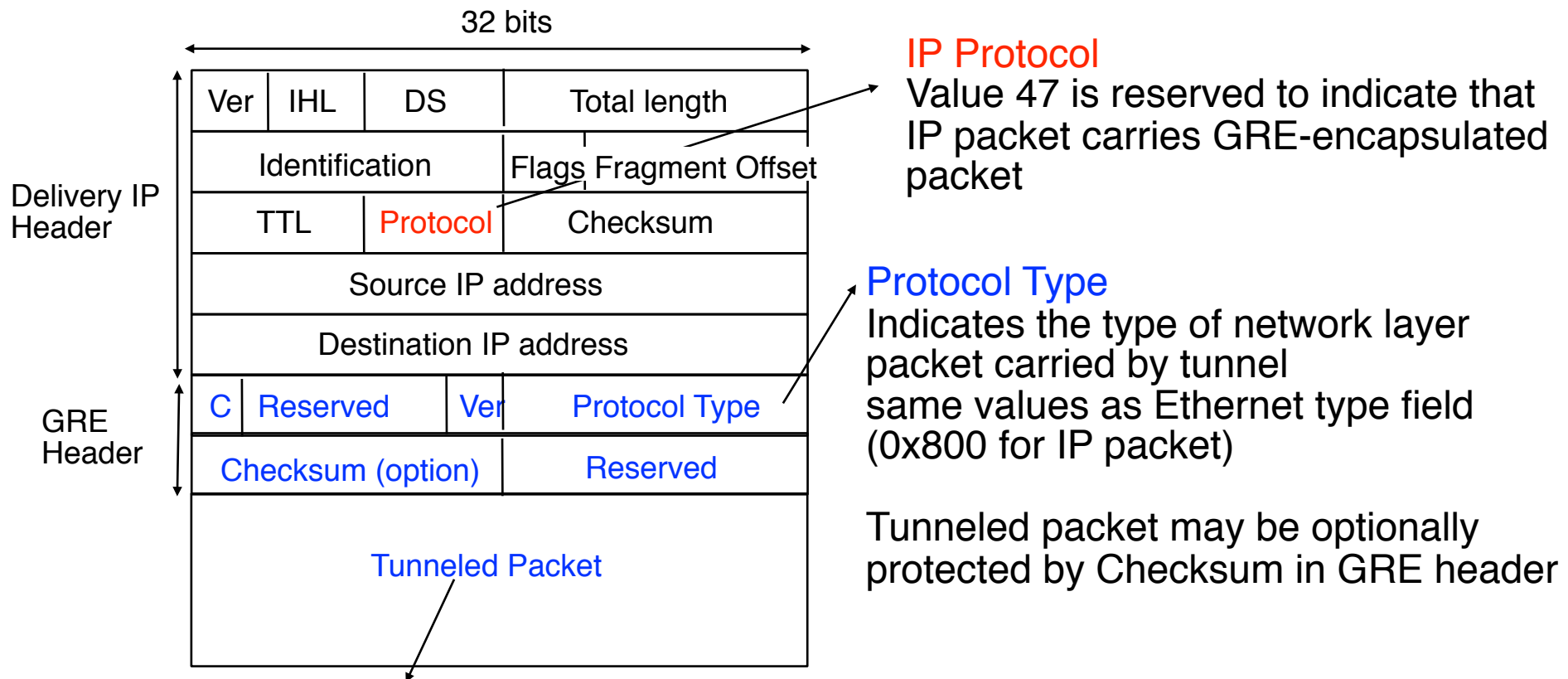


IP Tunnels

- Many IP tunnelling protocols exist
 - IP in IP tunnelling
 - can be used to carry IP packets inside IP packets
 - Generic Routing Encapsulation
 - can be used to carry network layer packets inside IP packets
 - Point-to-point tunnelling protocol
 - can be used to carry PPP frames through TCP/IP network
 - Layer 2 Tunnelling protocol
 - can be used to carry PPP frames through TCP/IP network
 - IPSec
 - security (authentication/confidentiality) extensions to IP also include tunnelling capabilities

GRE Tunnel

- Principle
 - Tunnel is used to carry network layer packets

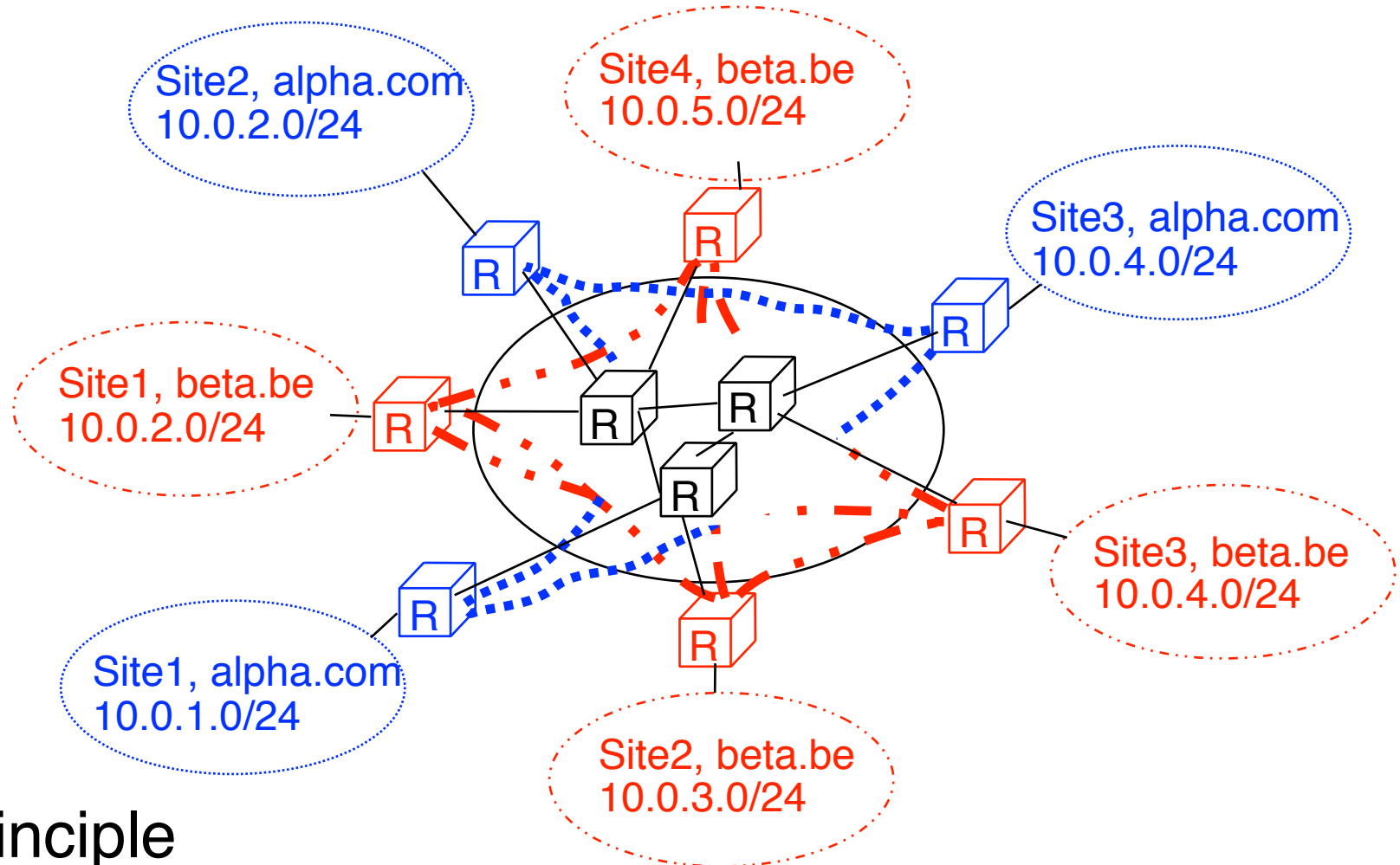


Can contain any network layer packet understood by destination system that can be placed inside Ethernet frame

Evaluation of the simple IP solution

- Advantage
 - Flexibility
 - a single physical interface on each router
 - Cost
 - VPN site can multiplex traffic to different sites on this link
- Drawbacks
 - the number of tunnels can be high
 - $n*(n-1)/2$ tunnels in total for full mesh
 - For a VPN with n sites, each router needs $n-1$ tunnels to obtain a full mesh
 - Flexibility
 - addition of a VPN require adding new tunnels
 - Security
 - depends on tunnelling mechanism used
 - weak with GRE, better with IPsec

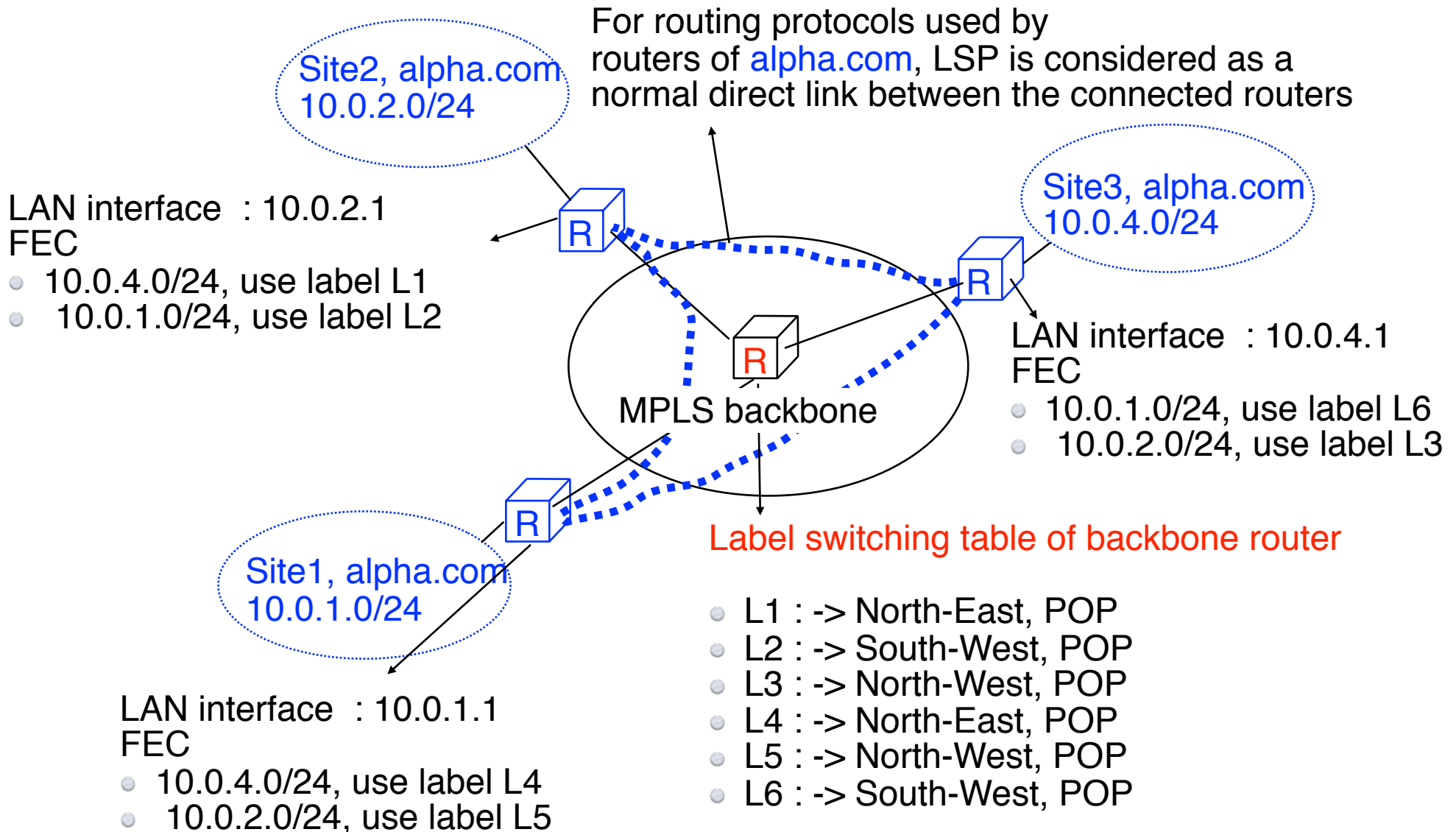
A simple MPLS-based solution



- Principle

- Manually create LSPs between customer routers from VPN sites through MPLS backbone

A simple MPLS-based solution (2)



Evaluation of the simple MPLS solution

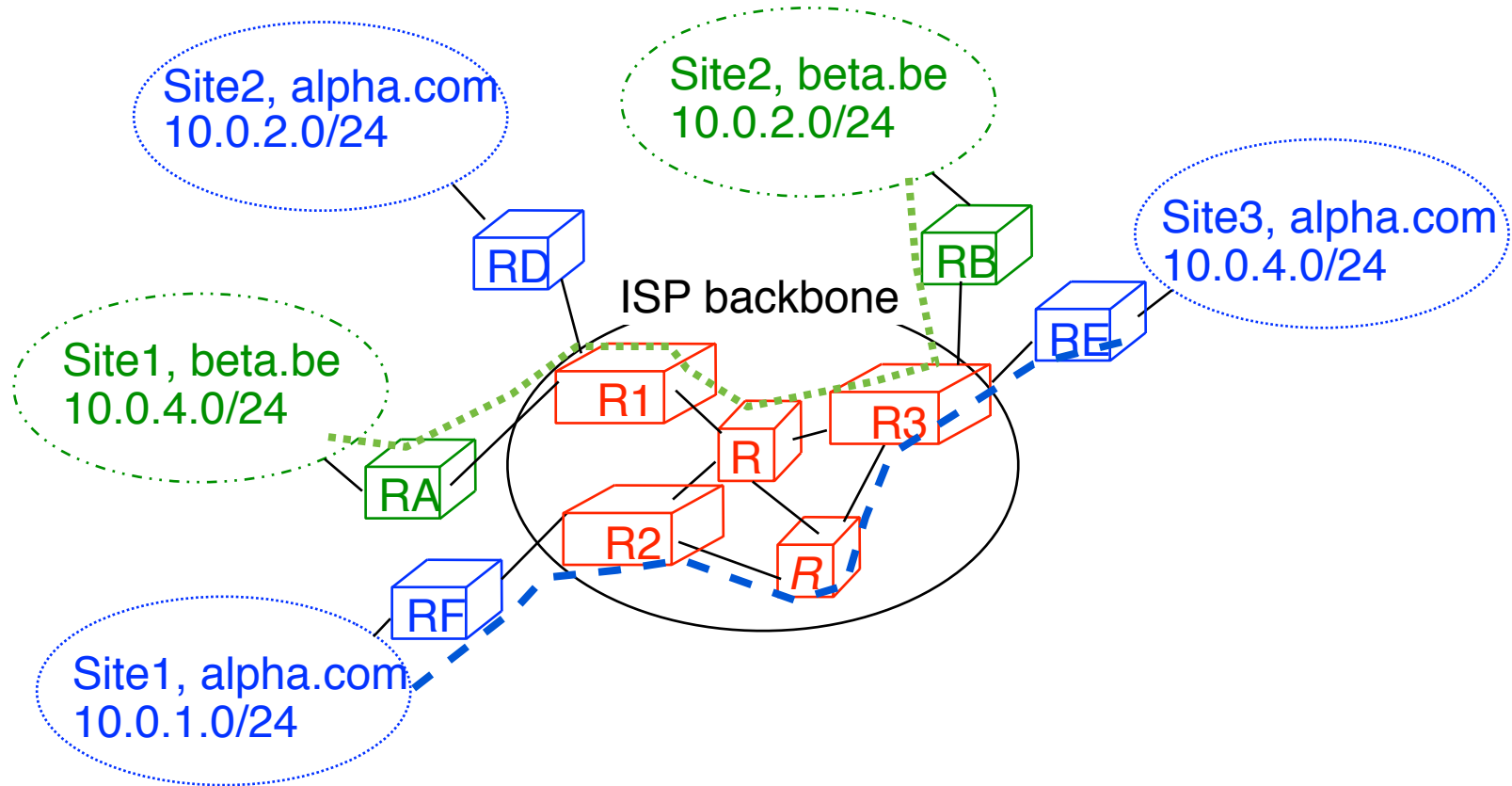
- Advantages
 - a single physical line per VPN site
 - QoS can be provided on a per-LSP basis
 - Flexibility
 - bandwidth of each LSP can be easily updated
 - Cost
 - statistical multiplexing is possible on MPLS backbone
- Drawbacks
 - MPLS support
 - routers of the VPN sites must support MPLS
 - backbone routers must support MPLS
 - configuration burden
 - backbone routers must be configured for each new LSP
 - customer routers must be configured for each new site

Provider-provisioned MPLS VPN

- Objective
 - Find a solution that is as automatic as possible
 - for the service provider
 - for the customers of the VPN service
 - Addition of a new site to an existing VPN
 - only the new customer router should need to be configured on the VPN
 - only a single router from the service provider should need to be configured on the provider's backbone

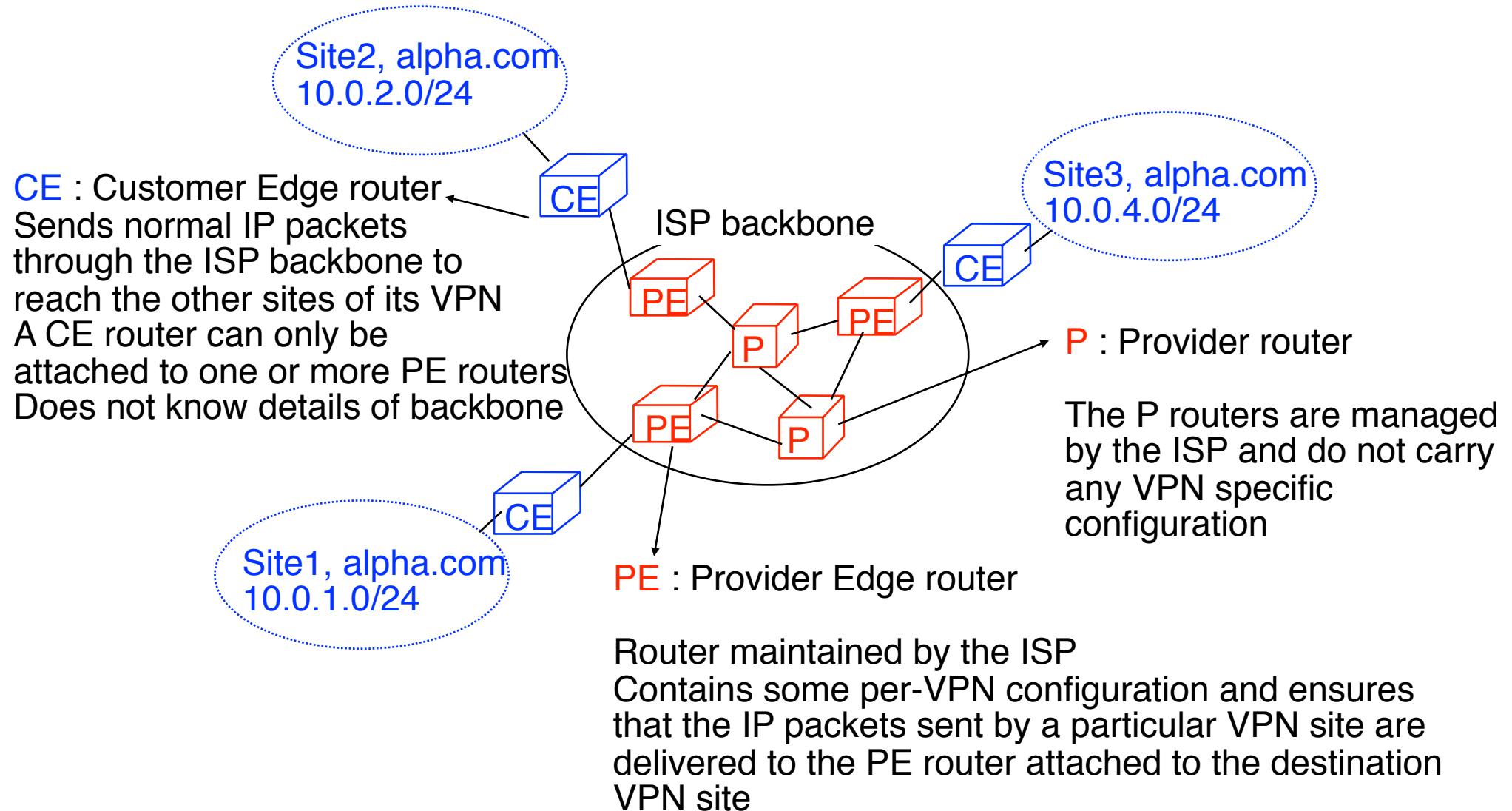
Provider-provisioned MPLS VPN (2)

- Principle of the solution

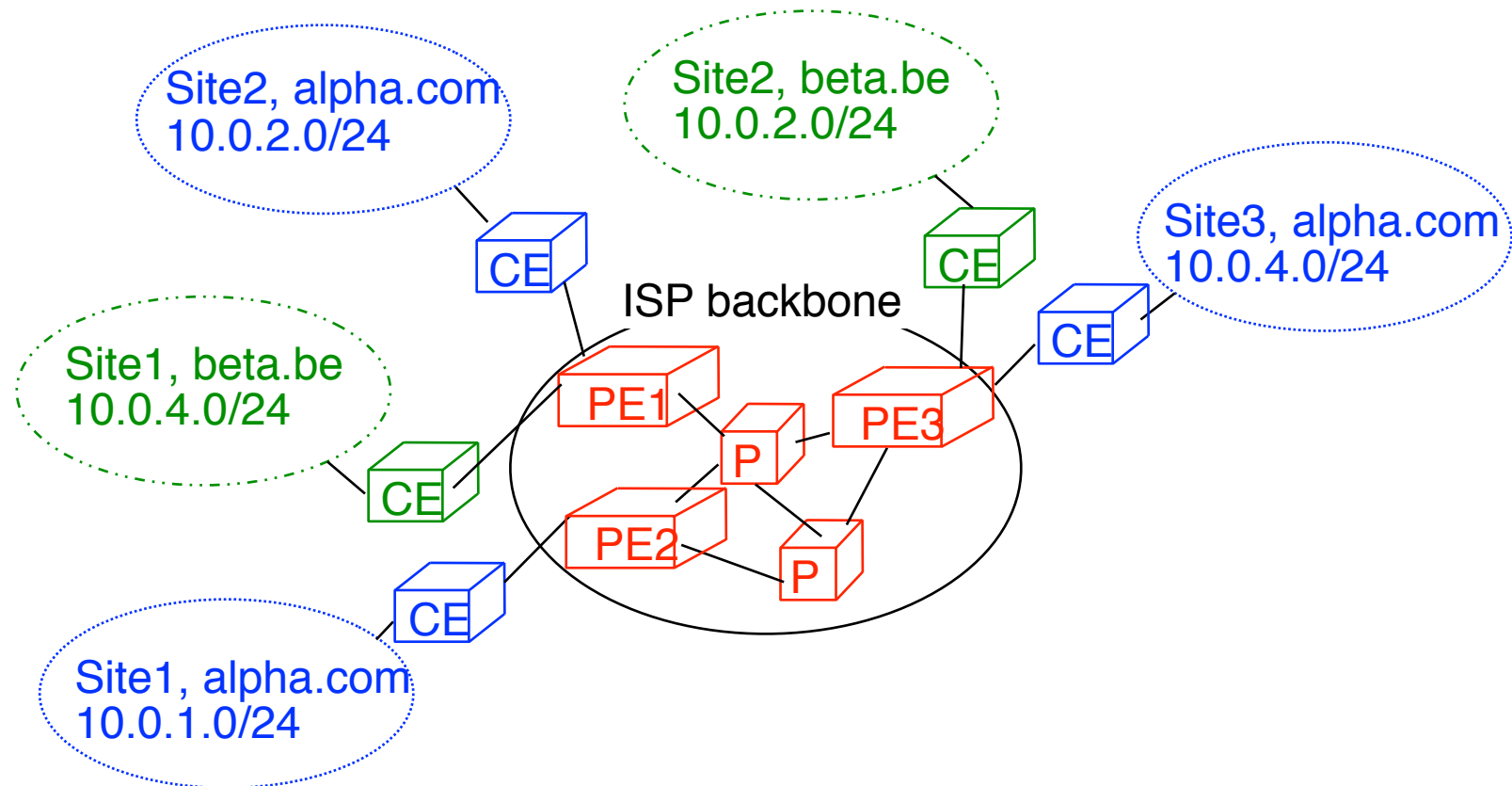


- transmission of one packet in **beta.be**, site1 to site2
- transmission of one packet in **alpha.com**, site1 to site3

Provider-based MPLS solution (3)



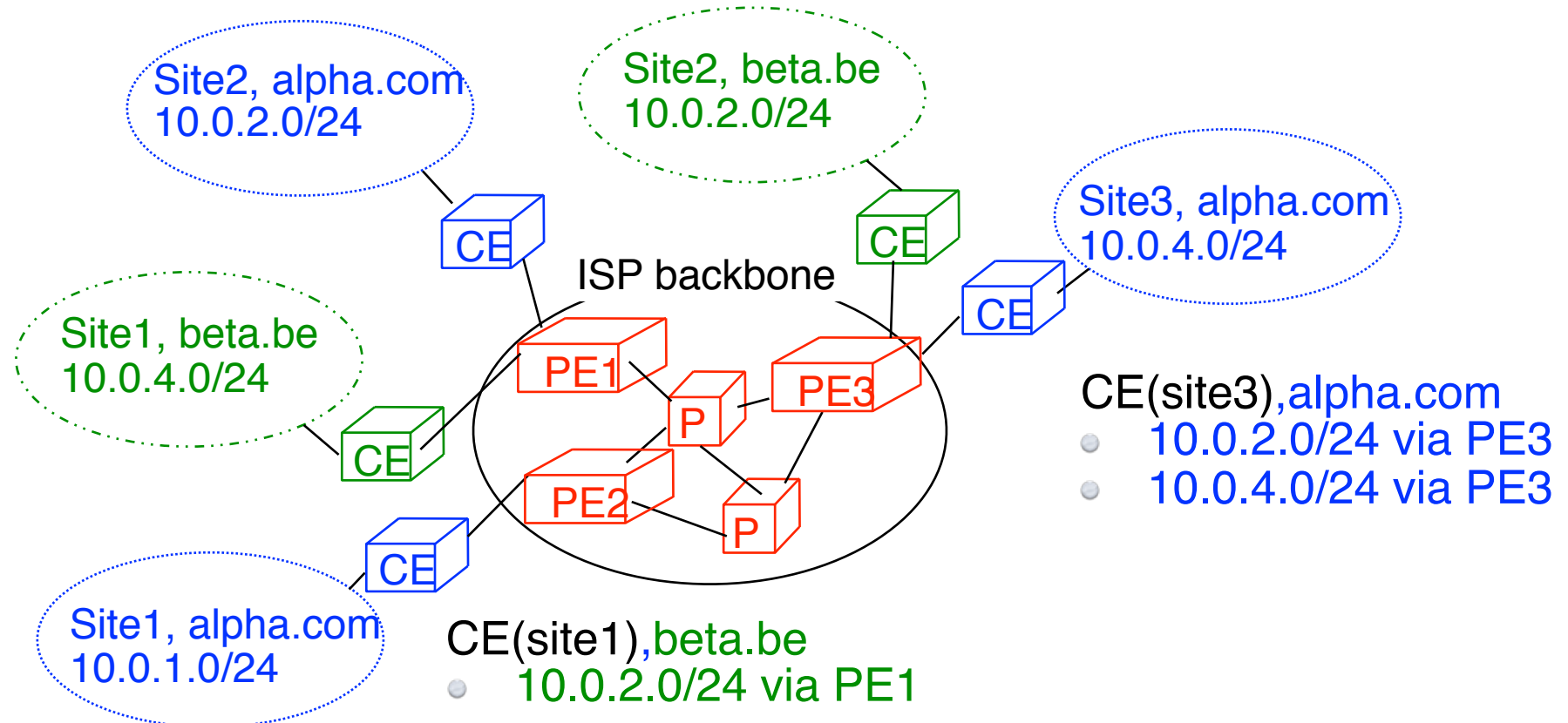
Problems to solve



- How to forward the packets from one CE router to the appropriate CE router of the same VPN ?
 - Need routing tables on CE, PE and P routers
 - How to efficiently distribute these routing tables ?

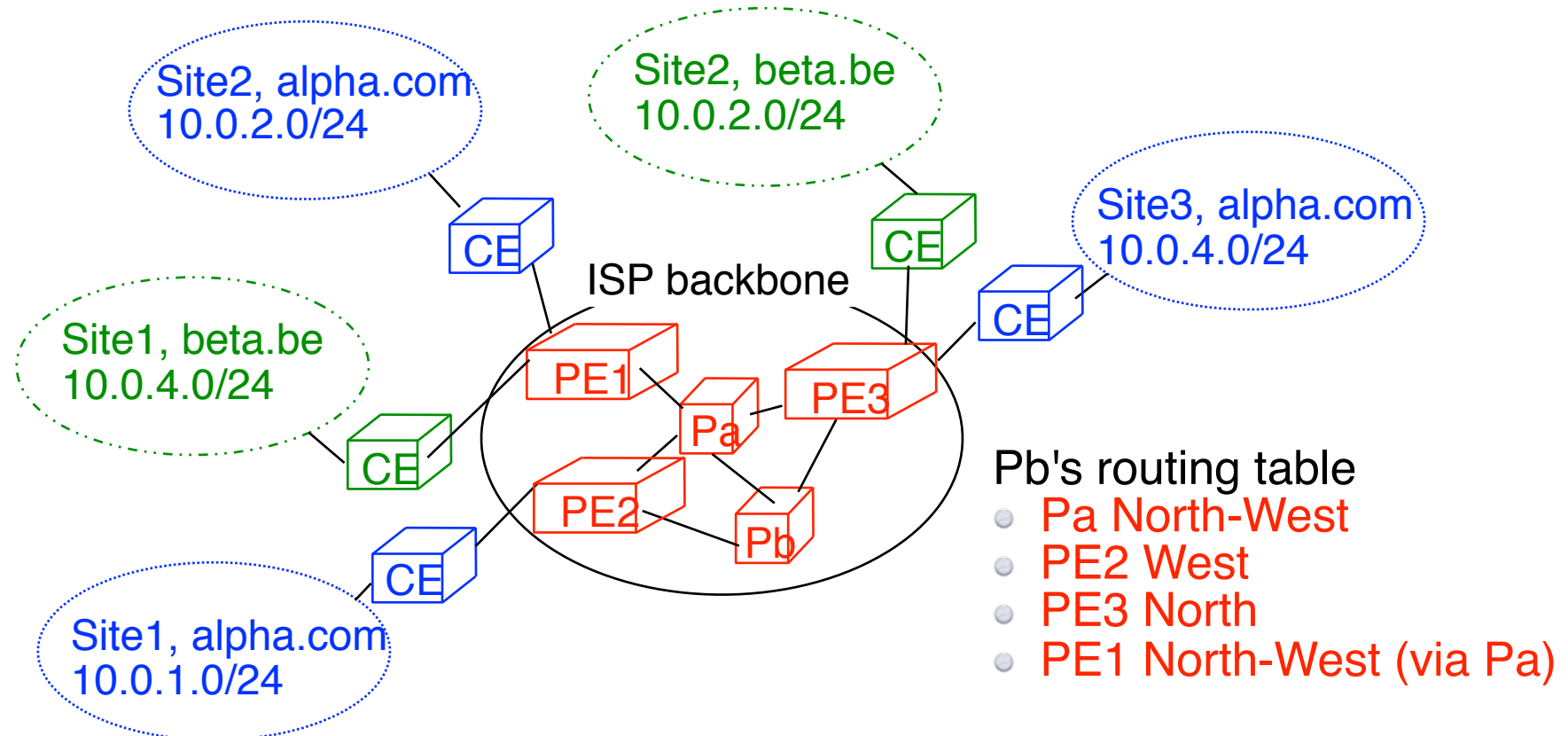
Routing tables on the CE routers

- Principle
 - Each CE router contains one routing table with the routes belonging to its VPN
 - CE does not know anything about ISP besides its attached PE



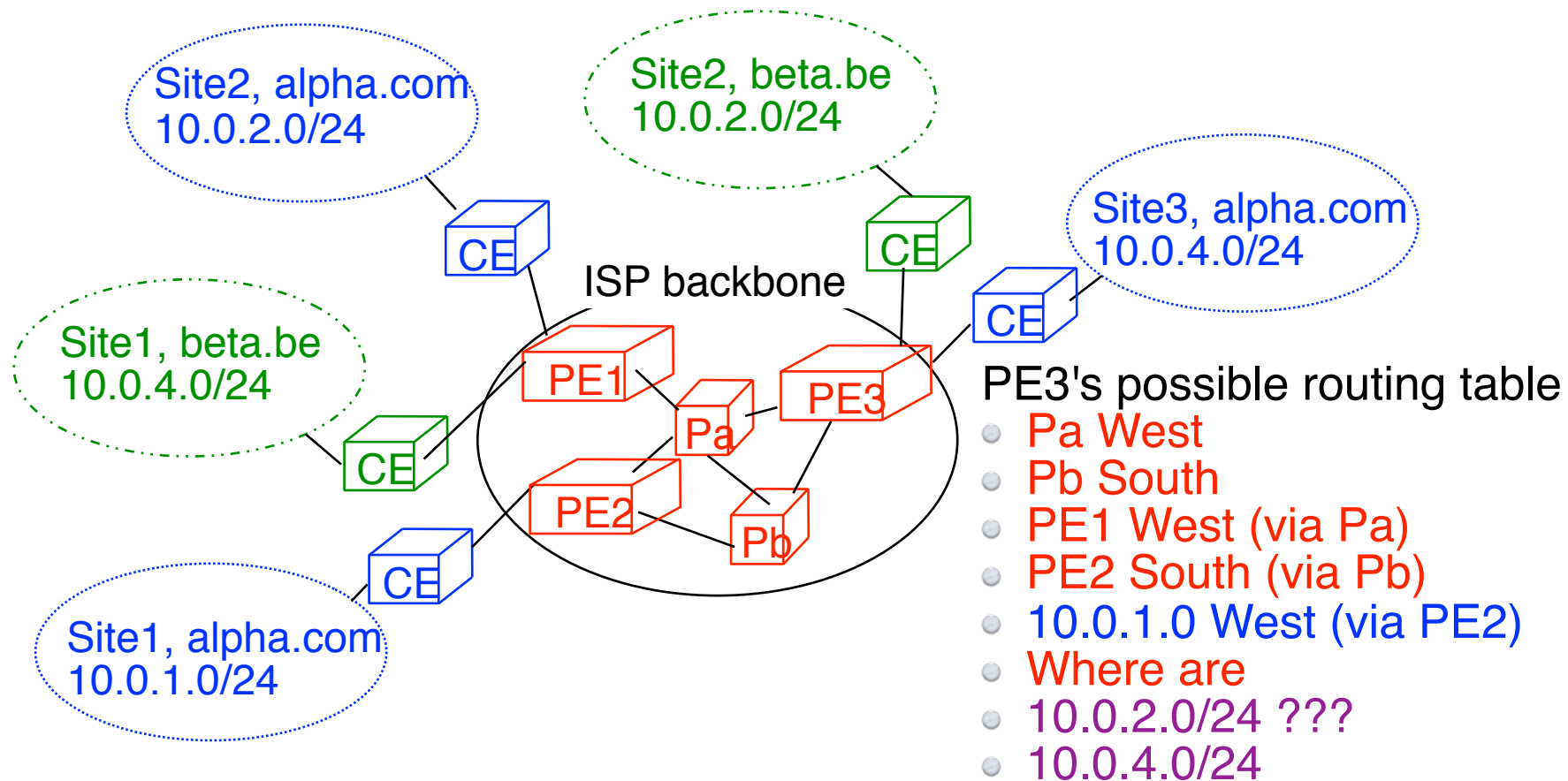
Routing tables on the P routers

- Principle
 - P routers only know how to reach the routers in their backbone
 - P routers do not know anything about VPNs



Routing tables on the PE routers

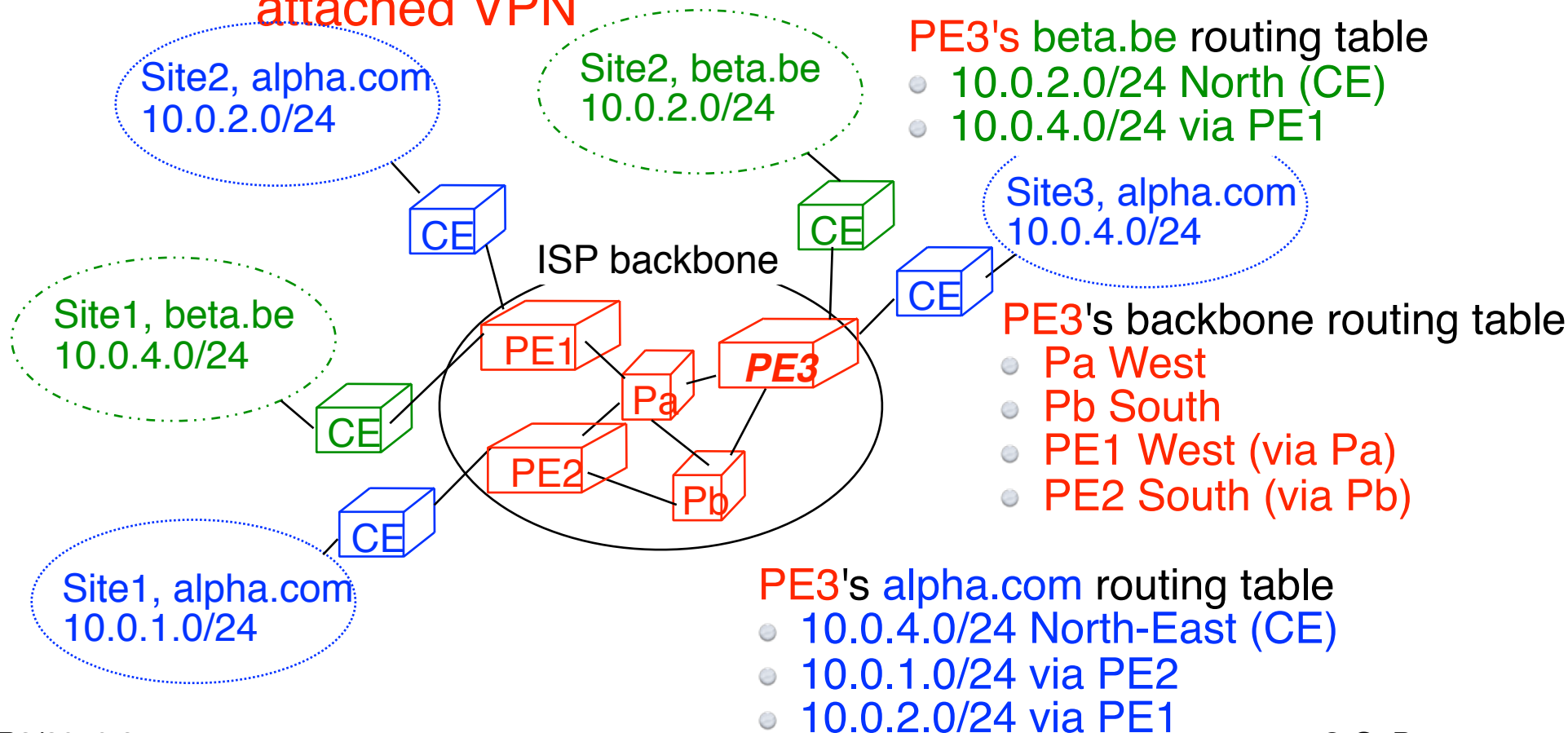
- Problem
 - Corporate networks often use RFC1918 addresses
 - Two different VPNs may use same IP subnets



Routing tables on PE routers (2)

- Principle

- Each PE router maintains several routing tables
 - standard routing table
 - one **VPN Routing and Forwarding table (VRF)** per attached VPN



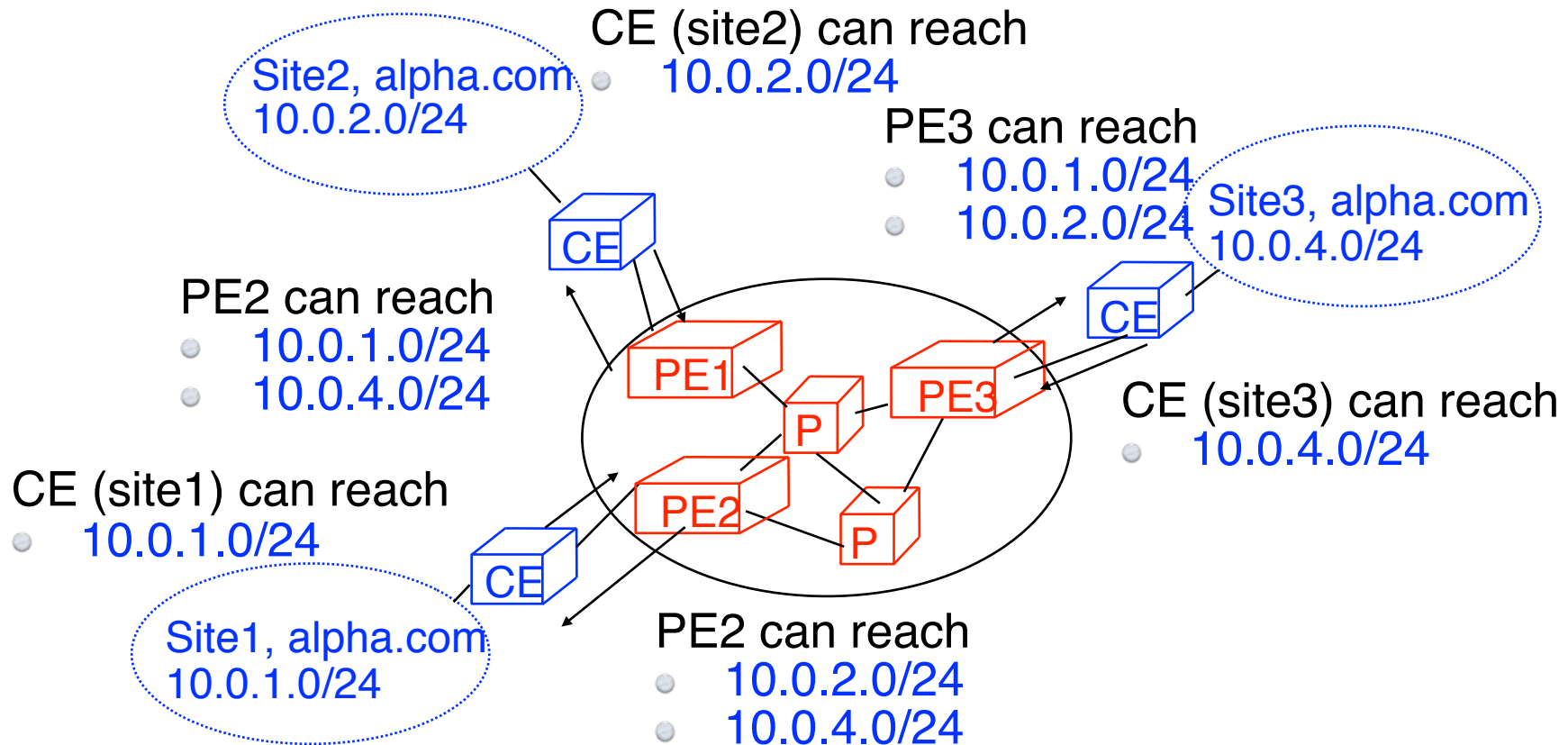
Distribution of the routing tables

- Routing problem
 - How can we correctly distribute the routing information to the CE, PE and P routers ?
 - A CE router must advertise its local routes to its attached PE and must receive the remote routes (or a default route) from this router
 - A PE router must receive two types of routing information
 - per VPN routing information for the routes reachable through attached CE routers and through remote PE routers
 - For scalability reasons, a PE router should only know the routing information about the VPNs that it directly supports
 - ISP routing information to be able to reach other PE routers
 - A P router must maintain routing information for the ISP
 - For scalability reasons, a P router should not know any VPN specific information

Distribution of routing information(2)

- Route distribution between CE and PE
 - static routes
 - both PE and CE are configured with static routes
 - suitable for small VPN sites with a single link
 - RIP
 - RIP is used by the CE to announce the routes reachable on its local network
 - RIP is used by the PE to announce the routes of the same VPN learned from the other PE routers
 - useful for medium VPN sites with multiple links
 - Other routing protocols
 - OSPF
 - This is a special OSPF instance between PE and CE, not the OSPF that is used inside the ISP backbone
 - eBGP
 - CE router uses eBGP session to advertise routes to PE

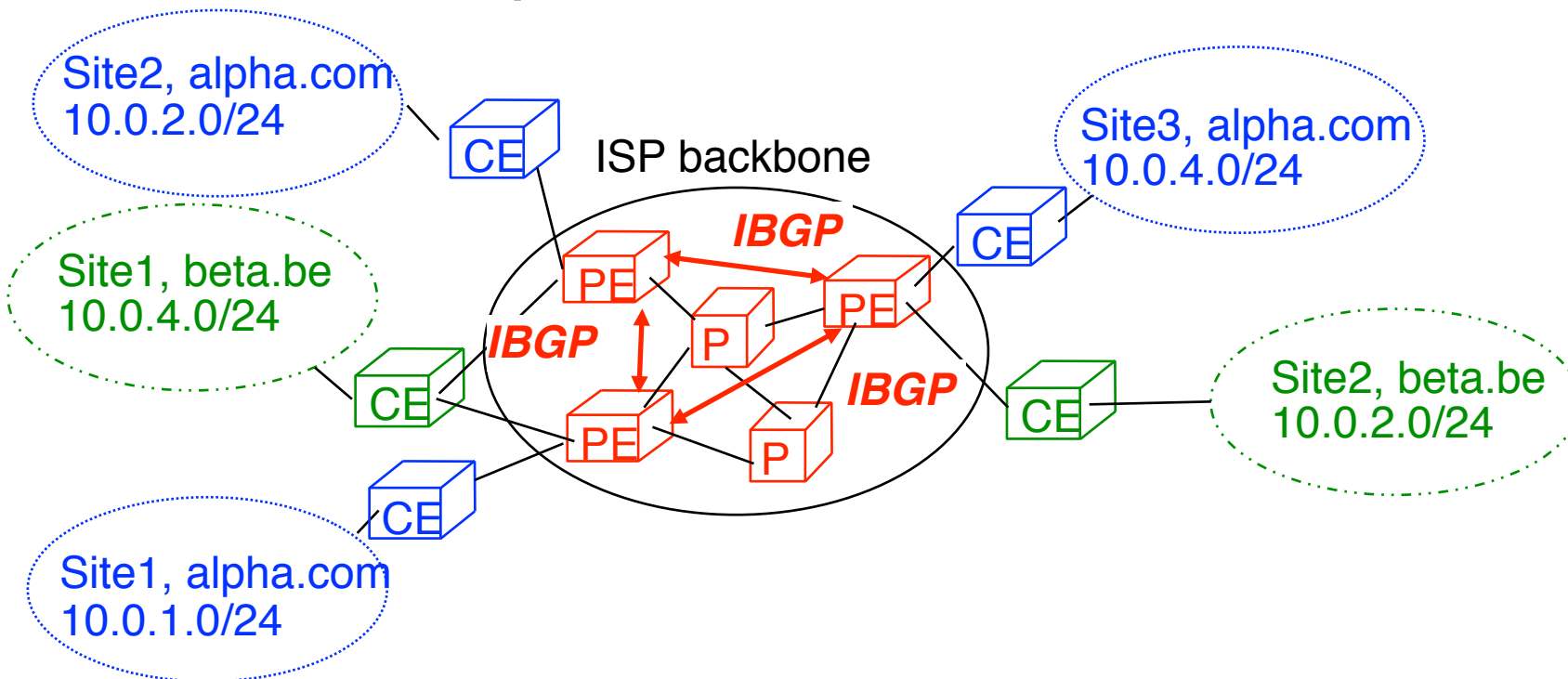
Distribution of routing information(3)



- In the backbone, all P and PE routers know the backbone topology by using OSPF

Distribution of routing information (4)

- Distribution of per VPN routes between PE

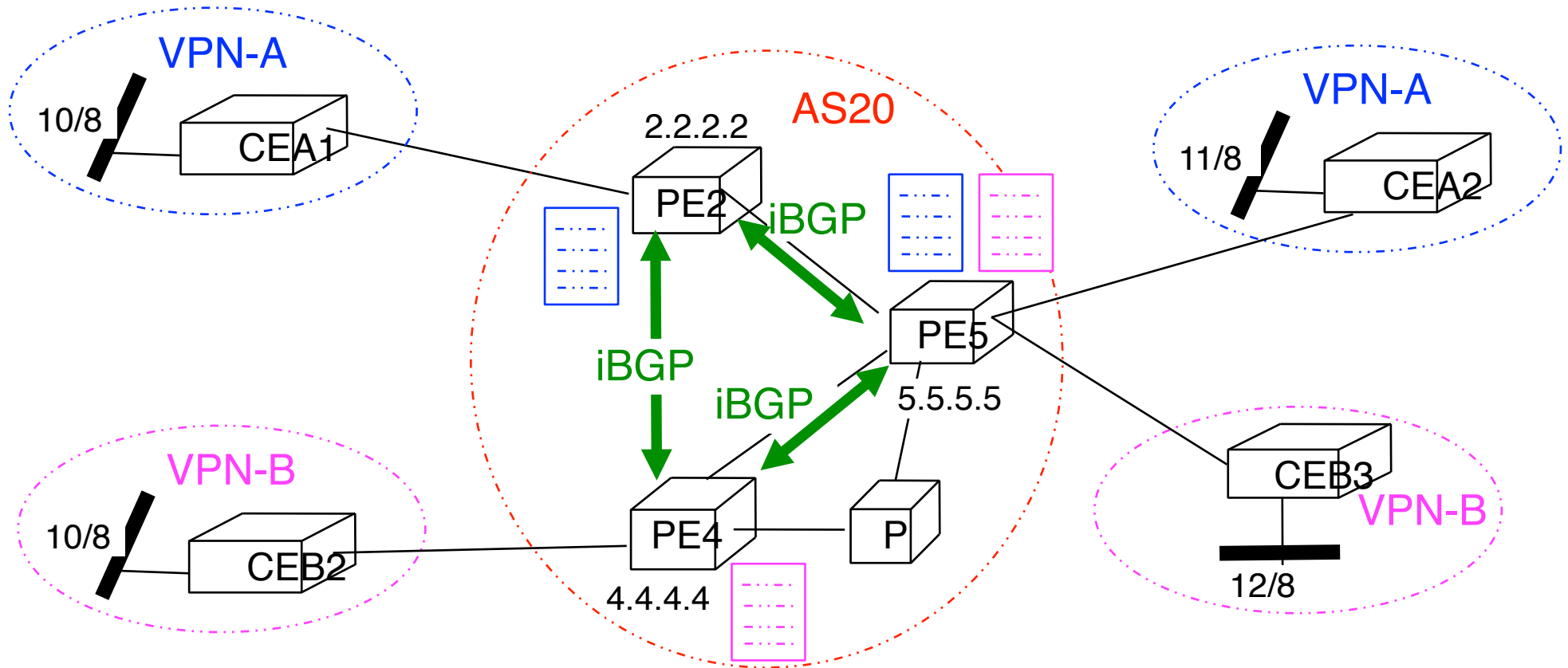


- Principle

- iBGP full mesh between PE routers
 - P routers do not need to run iBGP since they do not maintain per-VPN routes
- iBGP sessions are used to redistribute the routes learned from CE routers to distant PE routers

The distribution of the VPN routes by the PE routers

- Two issues
 - How to distribute the **A** and **B** routes for 10/8 ?
 - How to ensure that PE4 only receives **B** routes ?



MP-BGP and the VPN-IPv4 addresses

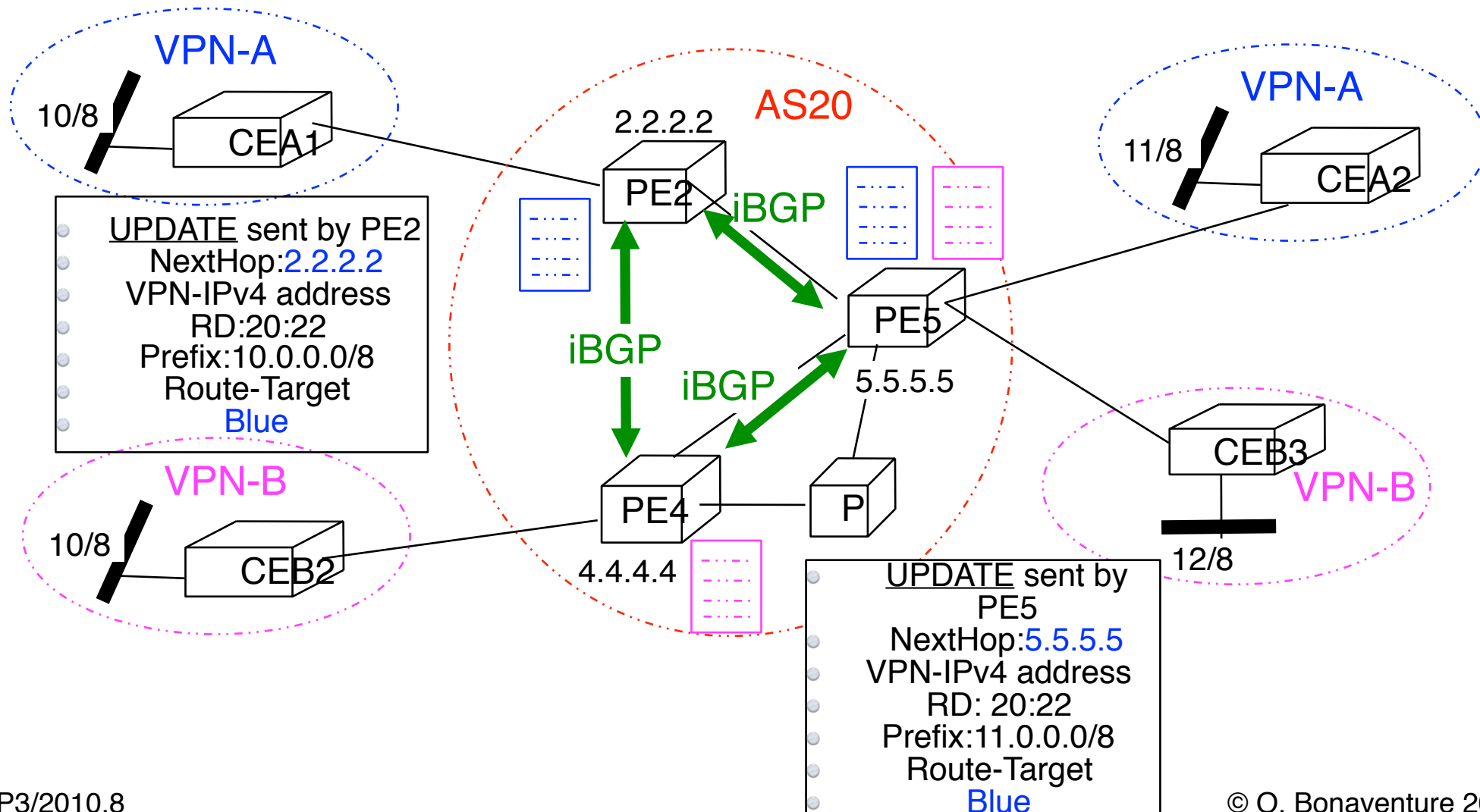
- MultiProtocol-BGP
 - an extension to BGP that allows a BGP router to advertise non-IPv4 routes
 - IPv6
 - IP multicast
 - VPN-IPv4
- The VPN-IPv4 address family
 - a method used by PE routers to encode IP v4 VPN addresses before advertising them with MP-BGP
 - a VPN-IPv4 address contains
 - an 8 bytes route distinguisher
 - an IPv4 prefix
 - BGP considers **VPN-IPv4 addresses** as *opaque bitstring*
 - two types of route distinguishers
 - AS:value
 - IPaddress:value

Controlling the distribution of VPN routes

- How to ensure that VPN-IPv4 routes only reach the PE routers attached to those VPNs ?
 - associate one or more **route targets** to each VRF
 - a route associated with RT x must be distributed to all PE routers that have a VRF with RT=x
 - RT is encoded as an BGP extended community
 - ASnumber:value
 - IPv4address:value
- Control of the distribution
 - PE router knows the RT supported by each of its peers and only advertises the appropriate VPN-IPv4 routes
 - or PE router advertises all its VPN-IPv4 routes and peers filter the received routes based on the attached RT

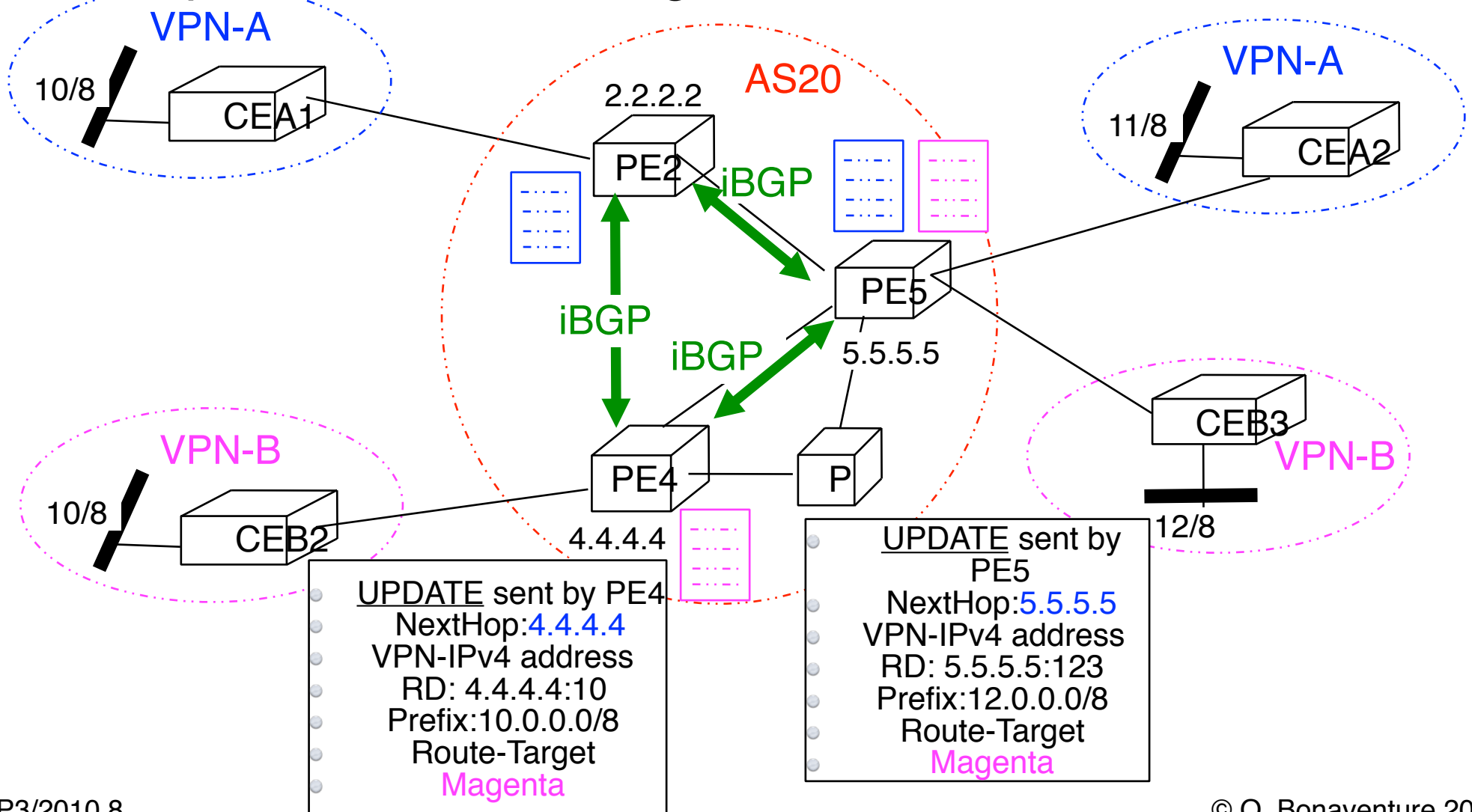
MP-BGP and the VPN-IPv4 addresses

- Example
 - per-VPN route distinguisher



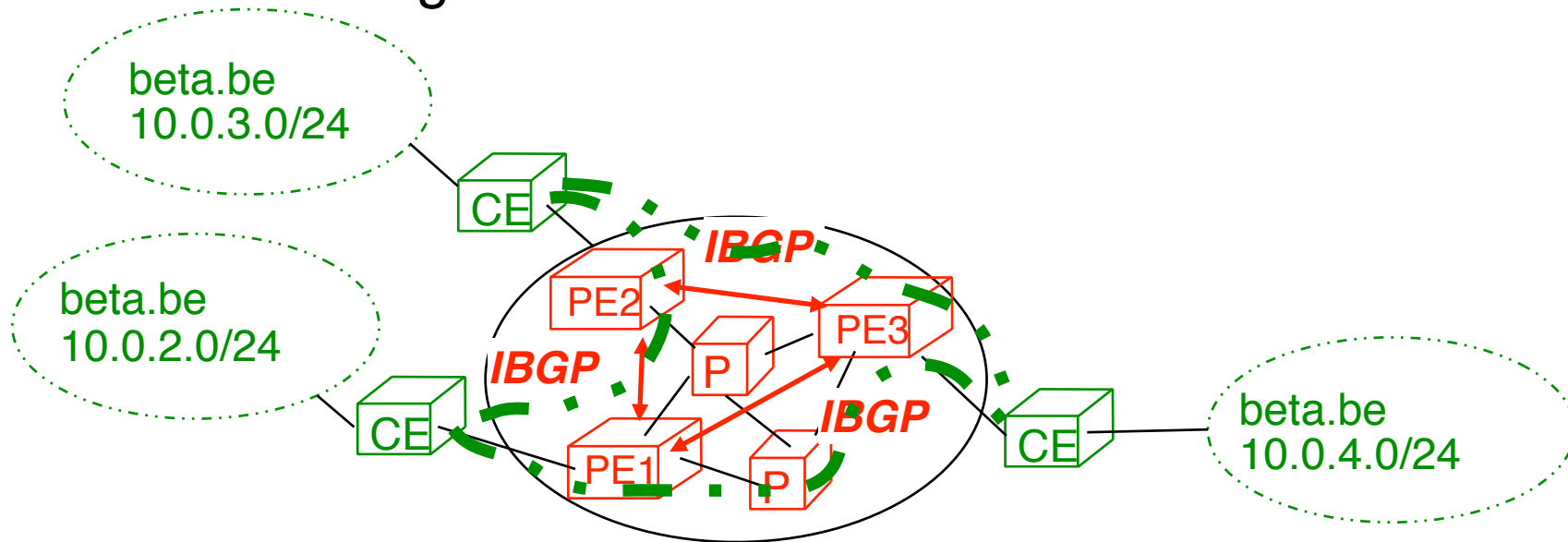
MP-BGP and the VPN-IPv4 addresses (2)

- Example
- per-site route distinguisher



Types of VPN connectivity

- Utilisation of the BGP extended community attribute
 - depends on the type of inter-sites connectivity within each supported VPN
- Full mesh connectivity
- all sites are equal
- one Route-Target for all sites of the VPN



Hub & spoke connectivity

-
- The diagram illustrates a multi-tenant network architecture. A central hub, labeled "Hub, beta.be 10.0.2.0/24", is connected to three spokes: "Spoke2, beta.be 10.0.3.0/24", "Spoke1, beta.be 10.0.4.0/24", and "Spoke3, beta.be 10.0.1.0/24". Each spoke is connected to the hub via a CE (Customer Edge) router. The hub and spokes are interconnected via PE (Provider Edge) routers. The diagram shows four PE routers (PE1, PE2, PE3, and PE4) and a central PE router. Red arrows labeled "IBGP" indicate internal BGP sessions between the PE routers. A dashed green line represents the network topology connecting the CE routers to the PE routers.

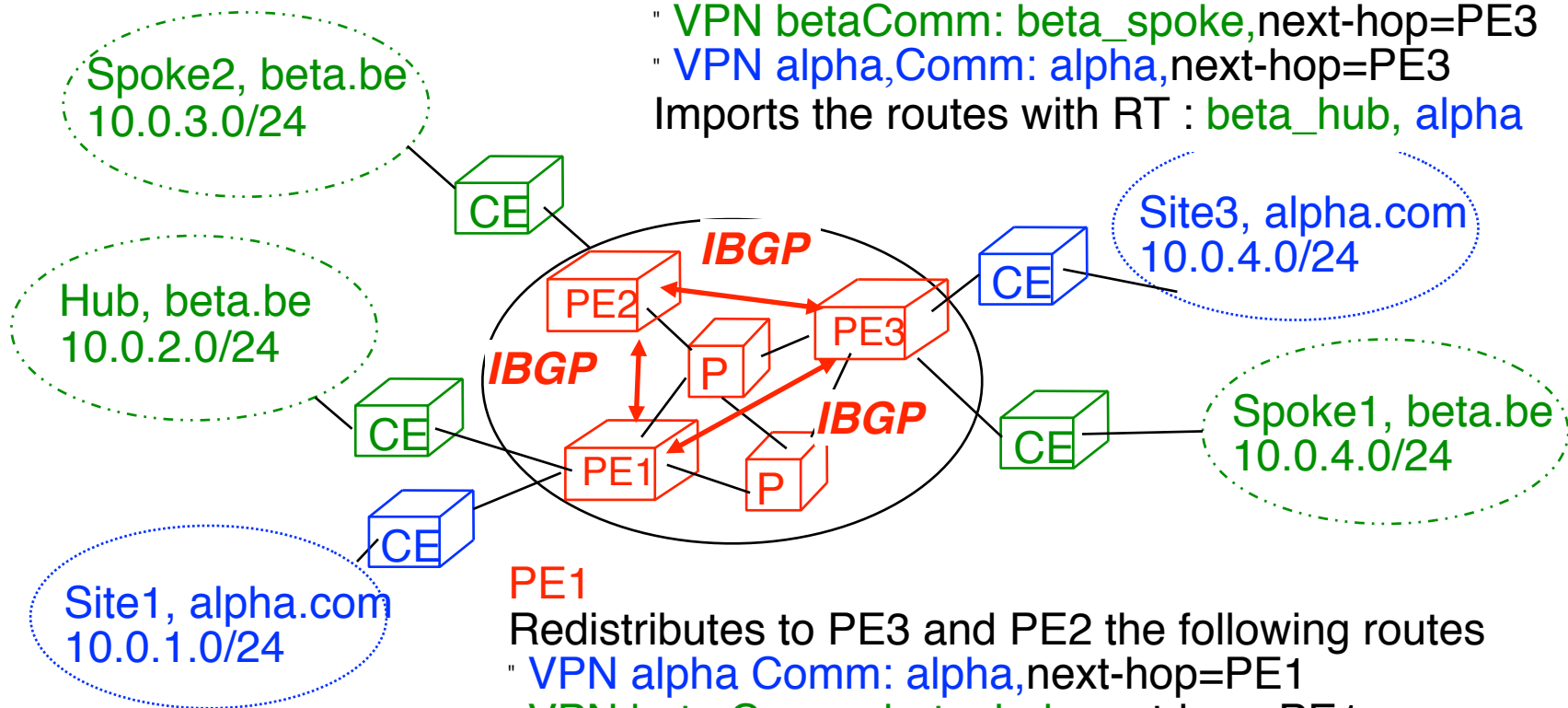
Types of VPN connectivity (3)

PE2

- "Redistributes : VPN beta Comm: beta_spoke,next-hop=PE2
- "Imports routes with RT : beta_hub

PE3

- "Redistributes
- " VPN betaComm: beta_spoke,next-hop=PE3
- " VPN alpha,Comm: alpha,next-hop=PE3
- Imports the routes with RT : beta_hub, alpha



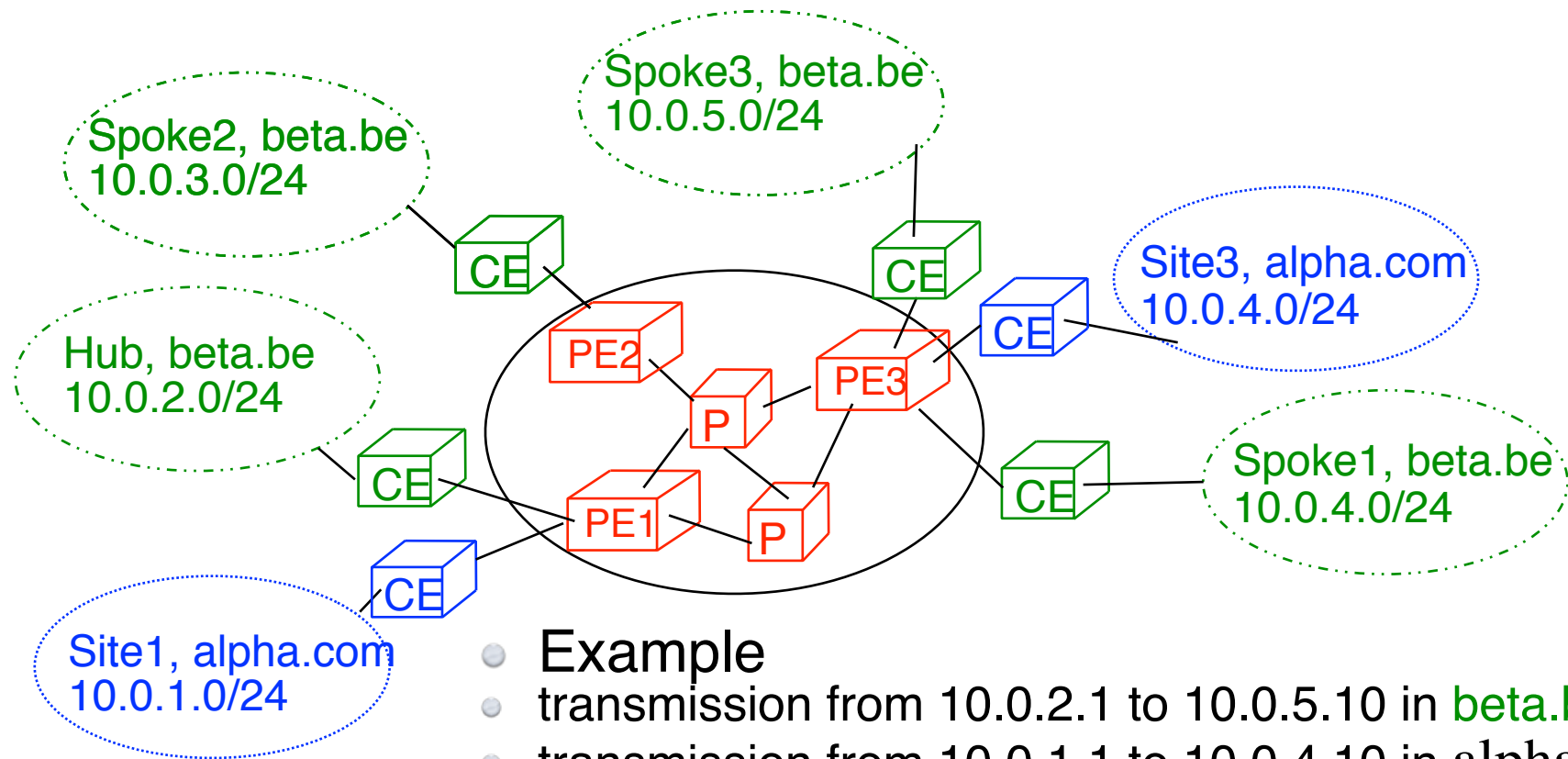
PE1

- Redistributes to PE3 and PE2 the following routes
- " VPN alpha Comm: alpha,next-hop=PE1
- " VPN beta Comm: beta_hub,next-hop=PE1
- Imports the routes with RT
- " alpha, beta_spoke

Solving the forwarding problem

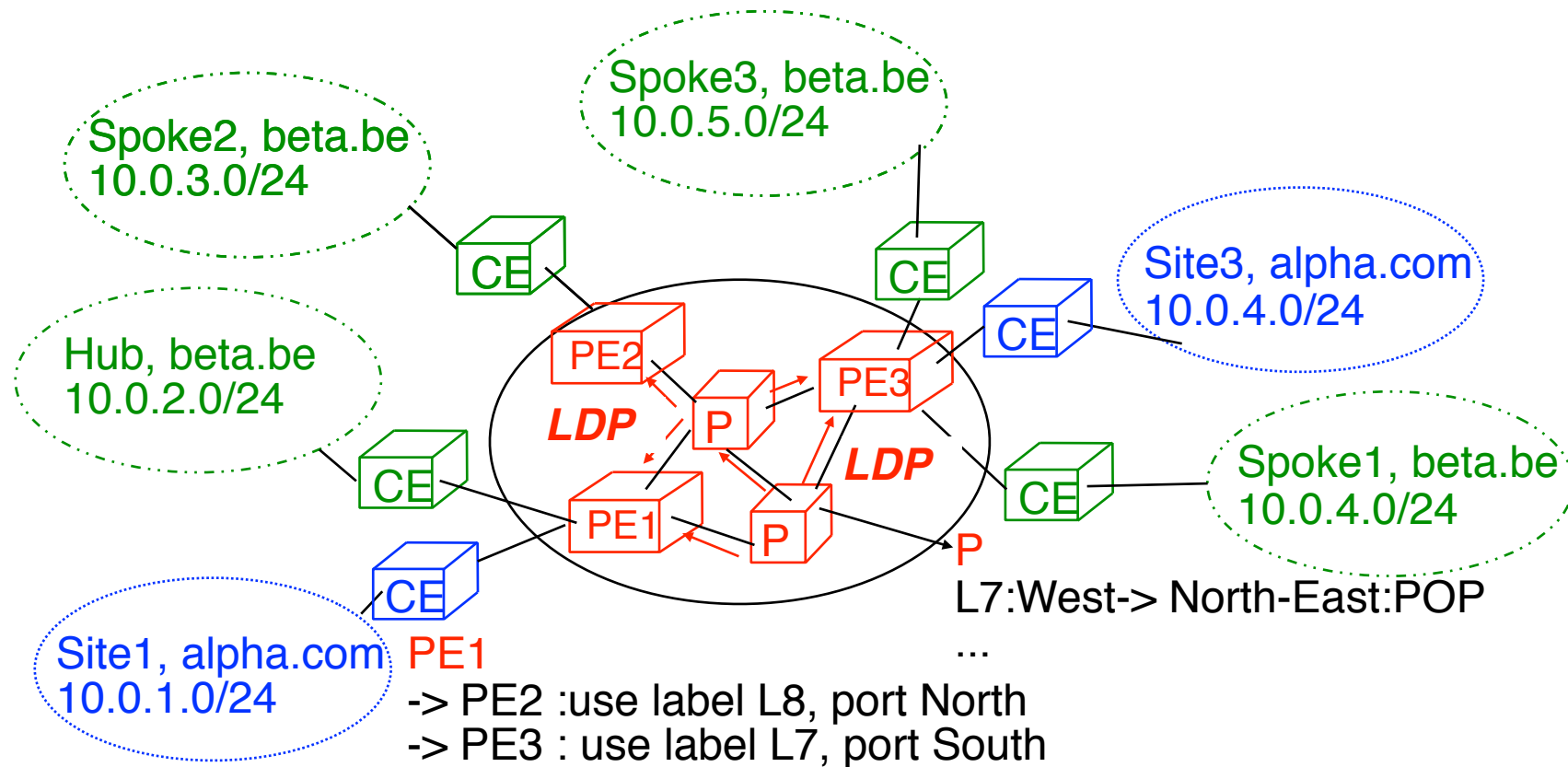
- How to forward the packets from each VPN through the provider's backbone ?
 - sending pure IP packets is not possible
 - P routers cannot know VPN-specific routes
 - different VPNs use the same RFC1918 address space
- Principle of the solution
 - CE routers send normal IP packets
 - CE routers remain as simple as possible
 - PE routers maintain **several** routing tables
 - **one routing table per VPN attached to PE router**
 - **one routing table for the ISP backbone**
 - PE encapsulate VPN packets
 - Common solution is to encapsulate with MPLS
 - Some ISPs are using GRE, L2TP or IPSec

Solving the forwarding problem with MPLS



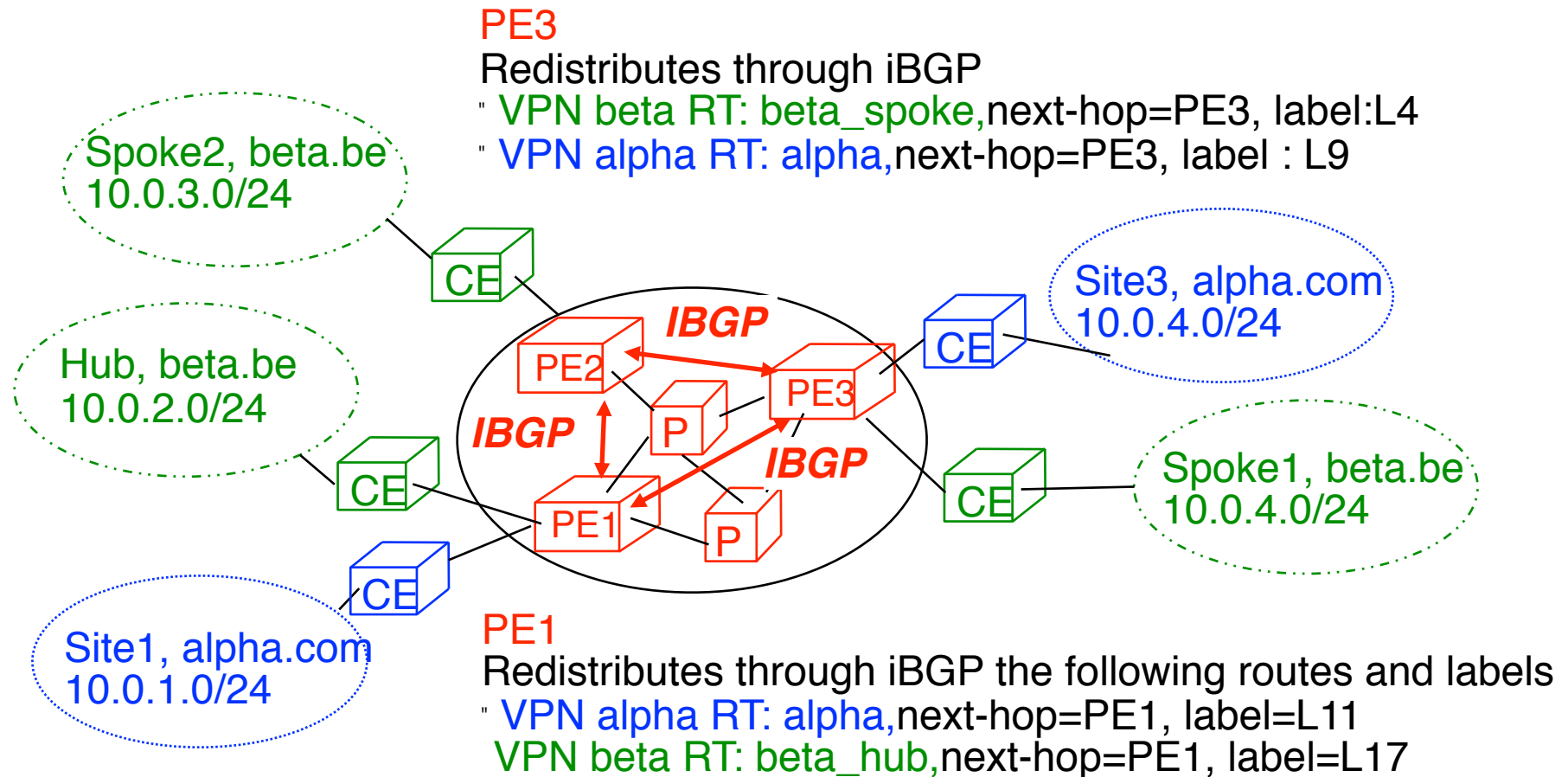
- Example
 - transmission from 10.0.2.1 to 10.0.5.10 in **beta.be**
 - transmission from 10.0.1.1 to 10.0.4.10 in alpha.com
- Principle of the solution : two levels of labels
 - one level of label is used to reach the next-hop PE
 - one level of label is used to indicate the VRF to be used (and thus the outgoing CE) in the egress PE

Distribution of labels



- Inside ISP backbone, use LDP to distribute labels between P and PE routers
 - each PE knows the label to use to reach any PE router
 - number of labels in P router depends on the number of PE, and not on the number of VPN sites

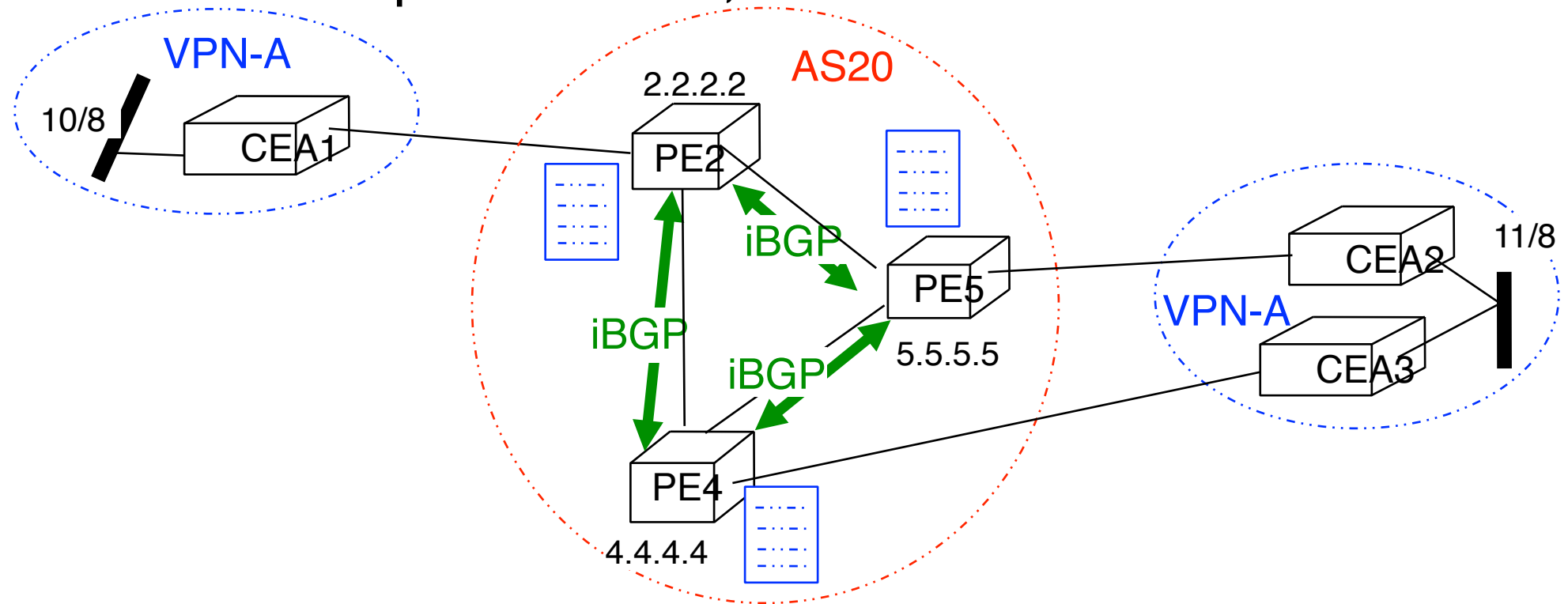
Distribution of labels (2)



- Principle
 - use iBGP to distribute VPN labels between PE routers

Packet flow in RFC2457 VPNs

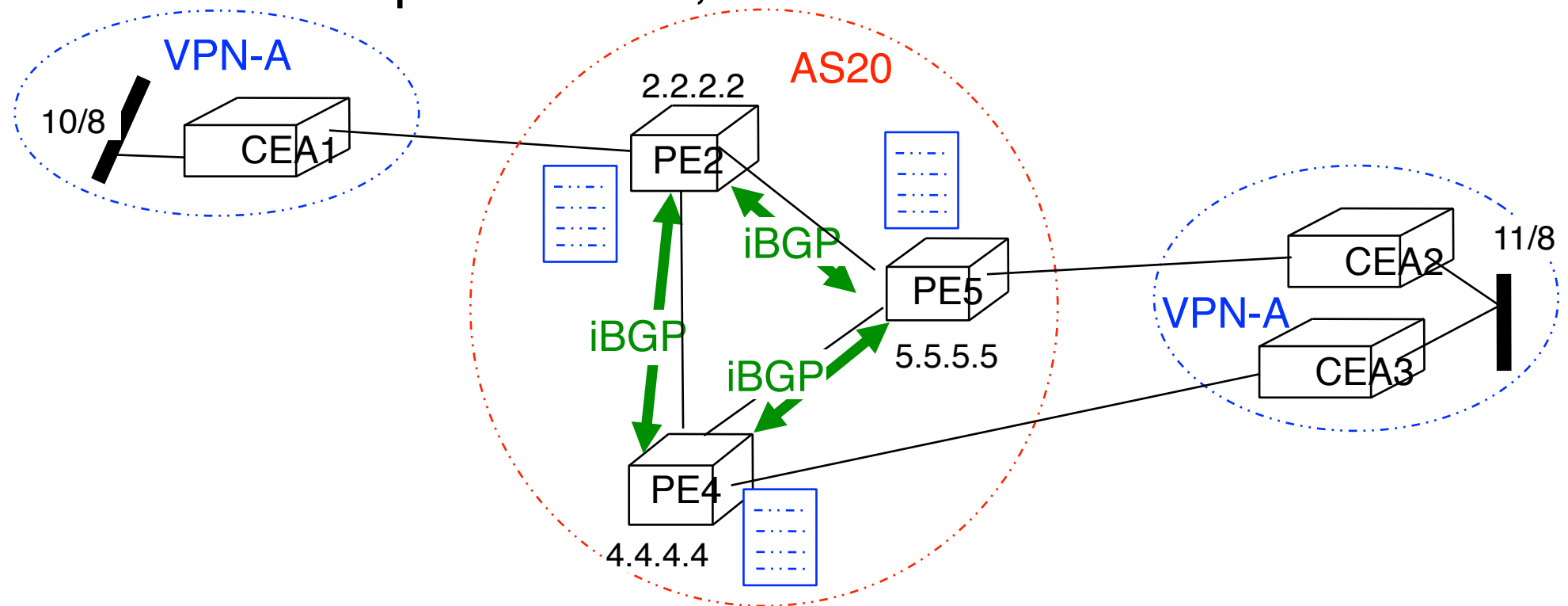
- with per-VPN RD, how does PE2 reach 11/8 ?



- PE2 receives two routes for 20:10:11/8
 - 20:10:11/8 from PE4 with nexthop = 4.4.4.4 (PE4)
 - 20:10:11/8 from PE5 with nexthop = 5.5.5.5 (PE5)
- PE2 selects the best route with its BGP decision process and installs it inside its **VPN-A** VRF
 - PE2 may use two LSPs to reach 11/8 via PE4 and PE5

Packet flow in RFC2457 VPNs (2)

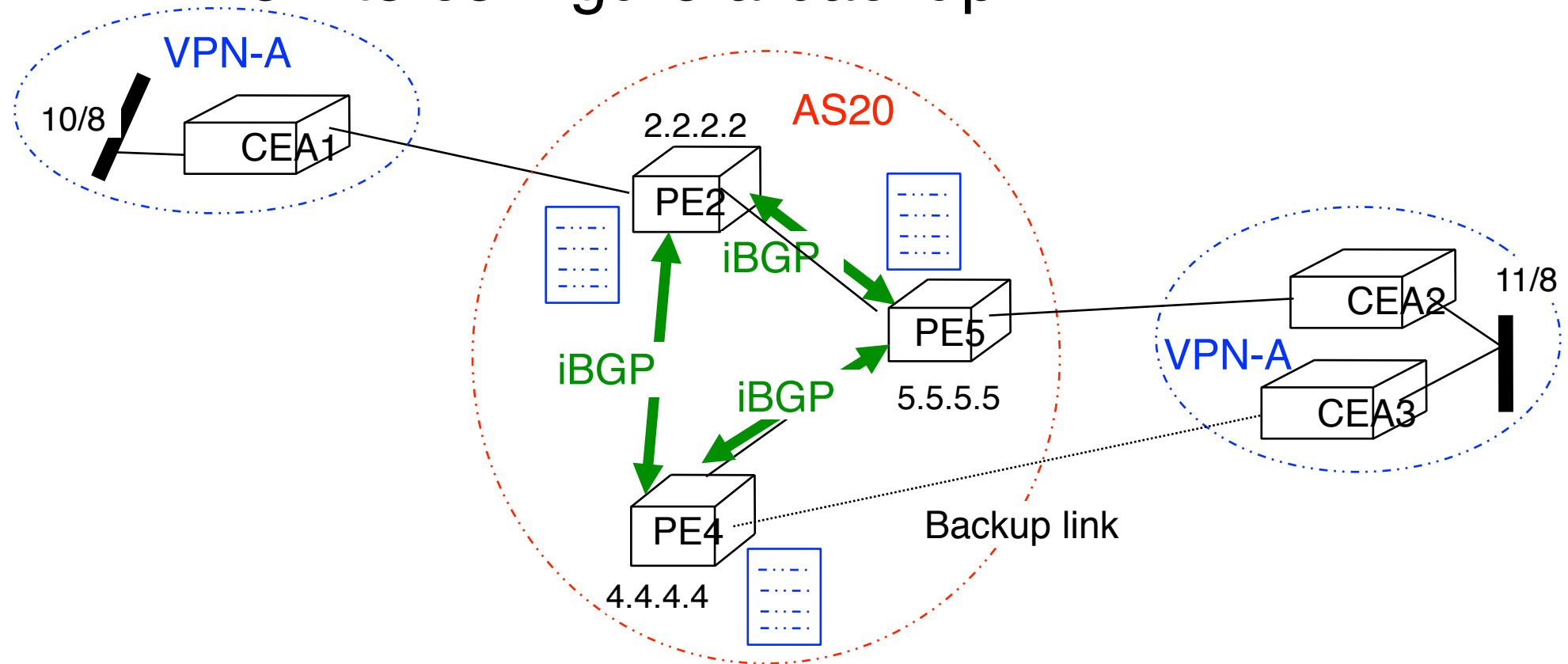
- with per-site RD, how does PE2 reach 11/8 ?



- PE2 receives two routes for 11/8
 - 4.4.4.4:123:11/8 from PE4 with nexthop = 4.4.4.4 (PE4)
 - 5.5.5.5:456:11/8 from PE5 with nexthop = 5.5.5.5 (PE5)
- BGP does not help PE2 to select which route is the best, the selection is done when installing in **VPN-A** VRF
 - PE2 may use two LSPs to reach 11/8 via PE4 and PE5

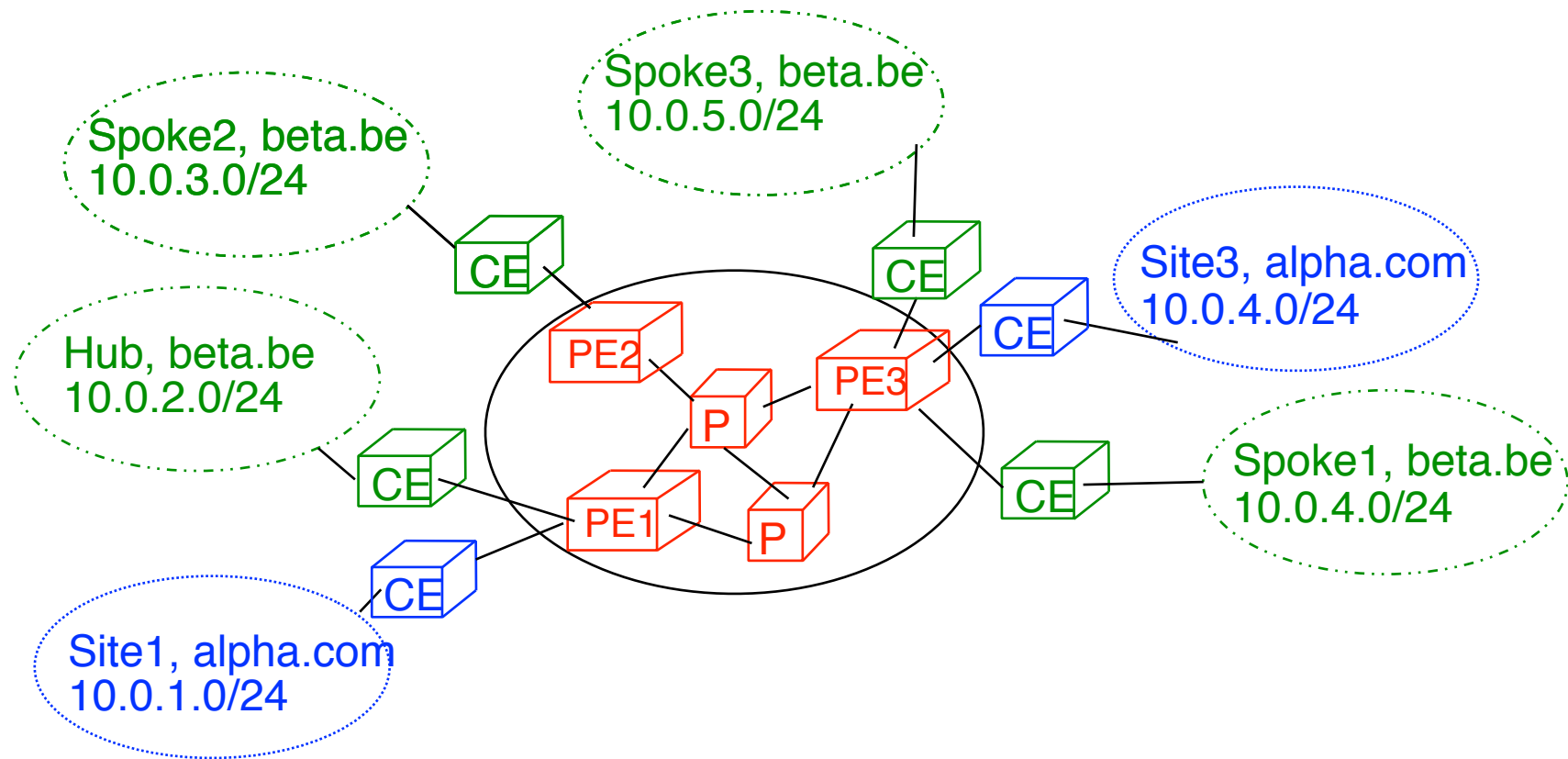
Backup links with RFC2457 VPNs

- How to configure a backup link ?



- PE4 adds localpref=50 to route learned from CEA3
- PE4 and all routers will prefer the route via PE5/CEA2
- Failure of link CEA2-PE5 will force PE5 to withdraw its VPN route towards 11/8 and the route via PE4 will be used

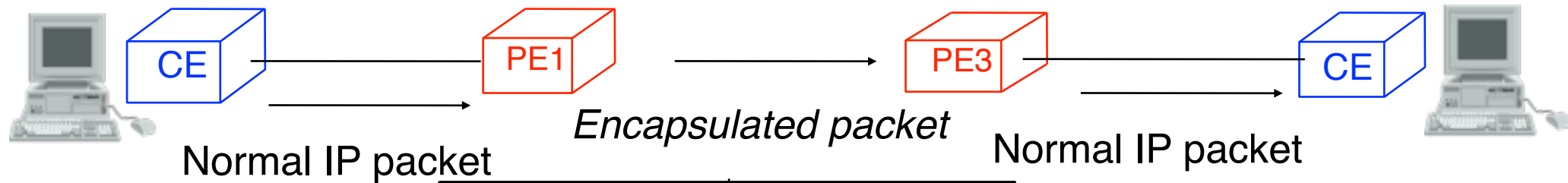
Solving the forwarding problem with tunnels



- Principle of the solution : Tunnel+MPLS
 - one tunnel is used to reach the next-hop PE
 - one MPLS label is used to indicate the VRF to be used (and thus the outgoing CE) in the egress PE

Solving the forwarding problem with tunnels (2)

- How to encapsulate the packets ?



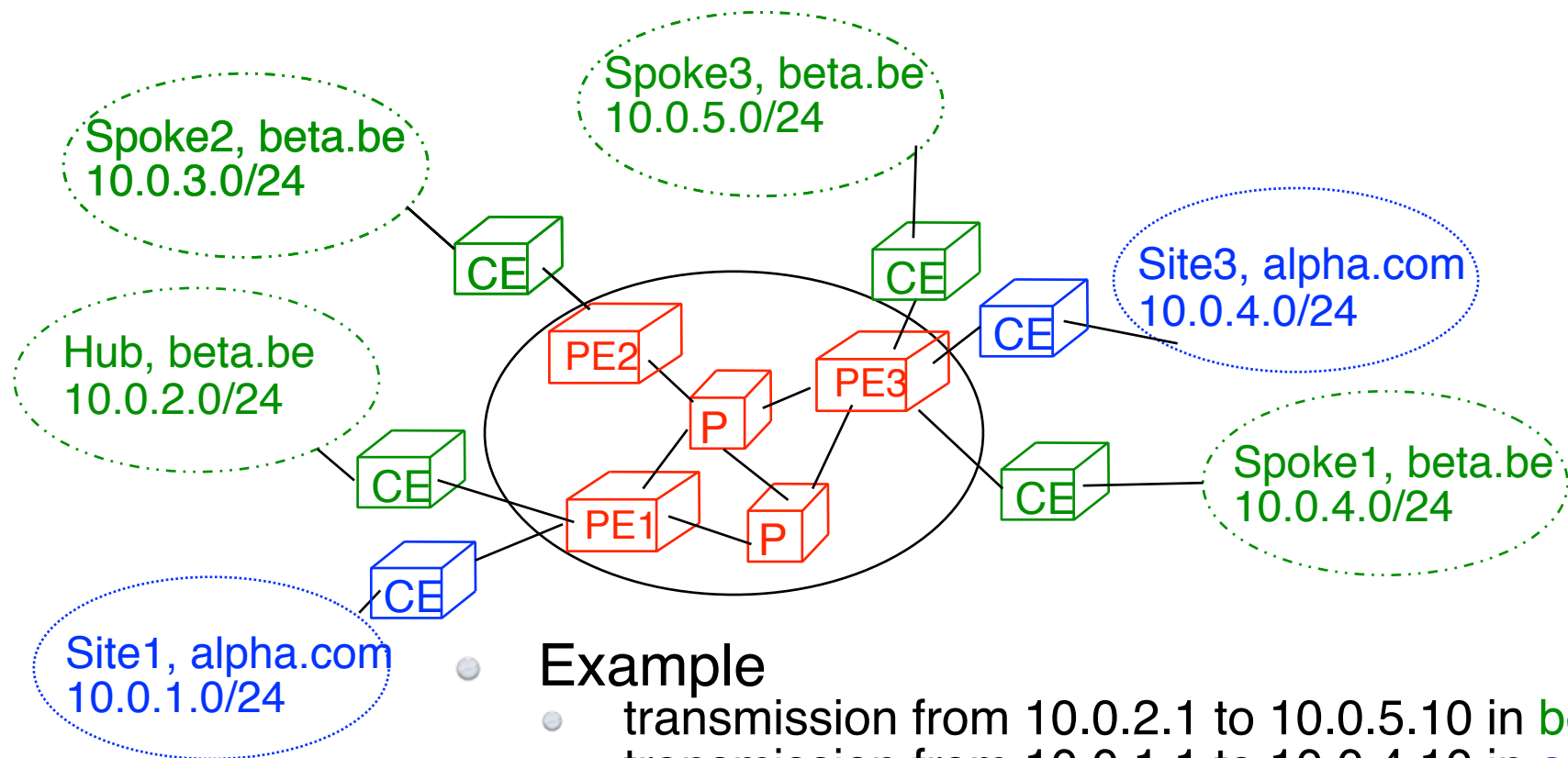
Ver	IHL	ToS	Total length		
Identification			Flags	Fragment Offset	
TTL	Prot.MPLS		Checksum		
PE1 IP address					
PE3 IP address					
MPLS Label				TTL	
Ver	IHL	ToS	Total length		
Identification			Flags	Fragment Offset	
TTL	Protocol		Checksum		
Source IP address					
Destination IP address					
Payload					

Solving the forwarding problem with tunnels (3)

PE3

Redistributes via iBGP

- "VPN beta RT: beta_spoke, next-hop=PE3, 10.0.5.0/24:label:L4
- "VPN beta RT: beta_spoke, next-hop=PE3, 10.0.4.0/24:label:L5
- "VPN alpha, RT: alpha, next-hop=PE3, label : 10.0.4.0/24L9



Example

- transmission from 10.0.2.1 to 10.0.5.10 in **beta.be**
- transmission from 10.0.1.1 to 10.0.4.10 in **alpha.com**

Comparison of VPN solutions

- Provider-provisioned BGP/MPLS VPNs
 - Easy to configure for customer and provider
 - Provider can provide special QoS to VPN
 - But customer routes are distributed inside the provider's network by iBGP
 - provider may need to carry a large number of routes if clients use /32, /30 or /28 subnets
 - some ISPs report BGP/MPLS VPN tables larger than the BGP tables of backbone Internet routers
 - stability and convergence time of routing in the customer network depends on provider's iBGP
 - BGP has a rather slow convergence
 - Customer does not entirely control routing in its VPN

Comparison of VPN solutions (2)

- Customer-provisioned VPNs
 - Providers are not involved in the provisioning of the VPN
 - no per-VPN routing tables to maintain and distribute
 - no revenue for value-added service
 - Customer builds VPN by establishing tunnels
 - it may be difficult to automate the tunnel establishment
 - a large number of tunnels may be required
 - Customer has full control over routing in the VPN
 - Routing protocol can be tuned for fast convergence, load balancing or whatever
 - no direct interactions between ISP's routing and VPN routing
 - Customer must be able to configure routers correctly

Layer 2 VPNs

- Service provided by RFC2457 VPNs is transport of IPv4 packets
 - CE devices are routers that send and receive IPv4 packets
- Some customers or operators prefer to offer layer 2 service
 - CE devices are capable of sending and receiving Ethernet frames, possibly with VLAN identifiers
 - Network managed by operator is similar to large Ethernet switch
 - PE devices need to learn MAC addresses reachable via each CE device
 - PE devices need to advertise to other PE devices the reachable MAC addresses
 - Broadcast and multicast Ethernet needs to be supported